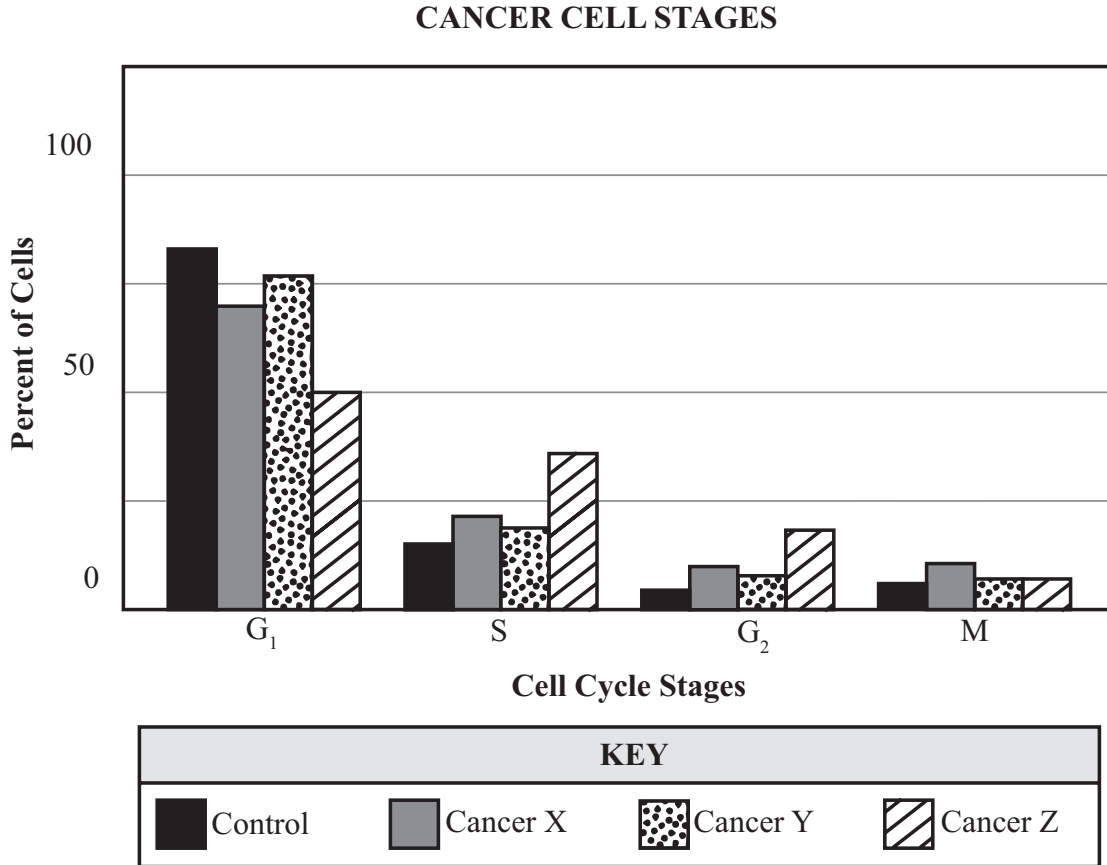




Biology 1 EOC Assessment

2026 Released Test Items

1. Cancer research involves the investigation of many cell factors. One factor is the stage of the cell cycle the cancer cells are in. A scientist conducted an investigation using various forms of cancer cells to compare the percentage of cells that are in each stage of the cell cycle. The graph below shows a summary of the data from one study.



Which analysis of these data can be made?

- Ⓐ The cancer cells spend less time in the G₂ stage than the control cells do.
- Ⓑ There is a greater percentage of Cancer Y cells than Cancer X cells in the M stage.
- Ⓒ There is a lower percentage of any type of cancer cells than the control cells in the G₁ stage.
- Ⓓ The control and Cancer Y cells spend more time in the S stage than Cancer X and Cancer Z cells do.

Biology 1

2. The process of meiosis is one way the cells of organisms prepare to divide. Which of the following is a characteristic of meiosis that results in genetic variation?
- Ⓐ division of somatic cells
 - Ⓑ formation of spindle fibers
 - Ⓒ replication of genetic material
 - Ⓓ crossing over of chromosomes

3. In 1995, scientists transferred a gene from wild rice into a type of cultivated rice. The expression of the transferred gene resulted in the cultivated rice being resistant to a specific bacterium that causes a deadly disease in rice plants. Which of the following **best** describes a benefit to society of this example of biotechnology?
- Ⓐ a reduction in the amount of nutrients removed from rice fields
 - Ⓑ a reduction in the need to apply fertilizers to rice fields
 - Ⓒ an increase in tolerance of rice for drought conditions
 - Ⓓ an increase in the production of rice for food

4. Three varieties of mangroves are found along the coastal regions of Florida. The red and the white mangroves inhabit the eastern and western coastal areas as far north as 29° latitude. The black mangrove overlaps with the other two varieties and grows even farther north, up to 30° latitude. Which of the following factors **most likely** contributes to the more northern aquatic range of the black mangrove?
- Ⓐ The black mangrove survives in colder water temperatures.
 - Ⓑ The black mangrove prefers the shallower coastal waters in the north.
 - Ⓒ The Gulf Stream provides nutrient-rich waters for the black mangrove.
 - Ⓓ The more northern waters provide the black mangrove with proper salinity levels.

5. A patient visits a physician because of a sore throat. The physician diagnoses the patient with an infection and determines that antibiotics will not be effective. Which of these is the **most likely** cause of this infection?
- Ⓐ bacteria
 - Ⓑ fungi
 - Ⓒ protozoans
 - Ⓓ viruses

Biology 1

6. In December of 1952, the amount of smoke and sulfur dioxide in the air of London, England, dramatically increased. A thick layer of smog covered the city as a result of a lack of wind and of an increase in burning coal to provide heat. The table below shows some of the data collected during the first half of the month.

AIR POLLUTANT LEVELS AND NUMBER OF DEATHS

Date (December 1952)	Sulfur Dioxide Levels (parts per billion)	Smoke Levels (micrograms/cubic meter)	Deaths per Day
1	98	257	257
3	148	445	327
5	375	1096	410
7	655	1595	905
9	452	1212	800
11	127	305	545
13	130	394	508
15	135	352	454

Which statement **best** describes the potential long-term effects on people who live in locations where these types of environmental conditions occur consistently throughout the year?

- Ⓐ Sulfur dioxide and smoke levels that rise quickly result in more respiratory illnesses than if the levels rise slowly.
- Ⓑ Repeated exposure to greater levels of smoke and sulfur dioxide may lead to an increased risk of respiratory problems.
- Ⓒ The elevation of sulfur dioxide levels occurs when smoke levels are also high, leading to illnesses in younger individuals.
- Ⓓ The most harmful level of pollutants affecting human health occurs when the amount of smoke and sulfur dioxide reach the same level.

7. In the 1830s, Matthias Schleiden examined the structure of plant cells and Theodor Schwann examined the structure of animal cells. After discussing their findings, they proposed an early version of the cell theory. Which of the following statements **best** describes why developing the cell theory was a scientific process?
- Ⓐ They created a hypothesis that was believed by other scientists.
 - Ⓑ They provided procedures that were approved by other scientists.
 - Ⓒ They made observations that could be repeated by other scientists.
 - Ⓓ They formed a conclusion that was immediately accepted by other scientists.

Biology 1

8. Students are asked to classify an unknown organism into the appropriate kingdom. The organism is multicellular with eukaryotic cells arranged into tissues. The cells of the organism do not have cell walls, and the organism is heterotrophic. In which of the following kingdoms should this organism **most likely** be placed?
- Ⓐ Animalia
 - Ⓑ Fungi
 - Ⓒ Plantae
 - Ⓓ Protista

9. Students are investigating the types of fish that live at different ocean depths. Which of the following describes characteristics of a fish that is **best** suited to live in deep ocean environments?
- Ⓐ a fish that is blue on the top and white on the bottom to reduce predation from above and below
 - Ⓑ a dark-colored fish that is adapted to cold water and uses a chemical light source to attract food
 - Ⓒ a blue-colored fish that feeds heavily on the types of algae that could damage a coral-reef system
 - Ⓓ a fish that relies on low-light conditions, on water with a high oxygen level, and on the use of a gas-filled chamber

Biology 1

10. A student read an article about the positive effects of extended exercise on the cardiovascular system. The student designed the investigation described below to test the claim that exercise decreased a person's blood pressure.

1. Record the initial blood pressure of 12 participants at noon on day 1 of the investigation.
2. On day 1, each participant exercises for 30 minutes by either walking on a treadmill or riding an exercise bicycle.
3. Each participant alternates the type of exercise performed for 30 minutes for six consecutive days, followed by one day of rest, for 12 weeks.
4. At noon on the seventh day of each week, record the blood pressure of each participant.
5. At the end of 12 weeks, calculate the change in blood pressure for each participant and calculate the average change in blood pressure for the group.

Which of the following would **most** improve the design of the student's investigation?

- Ⓐ Include participants that do not exercise for comparison.
- Ⓑ Have each participant conduct a different type of exercise.
- Ⓒ Allow some participants to exercise for longer periods of time.
- Ⓓ Measure blood pressure of the participants during exercise each day.

Fertilizer Runoff and Water Quality

Students in Ms. Macedo's class know that fertilizers add nitrogen-, phosphorus-, and potassium-containing compounds to the soil. They decided to conduct an investigation to determine the effects of fertilizer runoff on the water quality of a river that runs through a wooded area near a golf course.

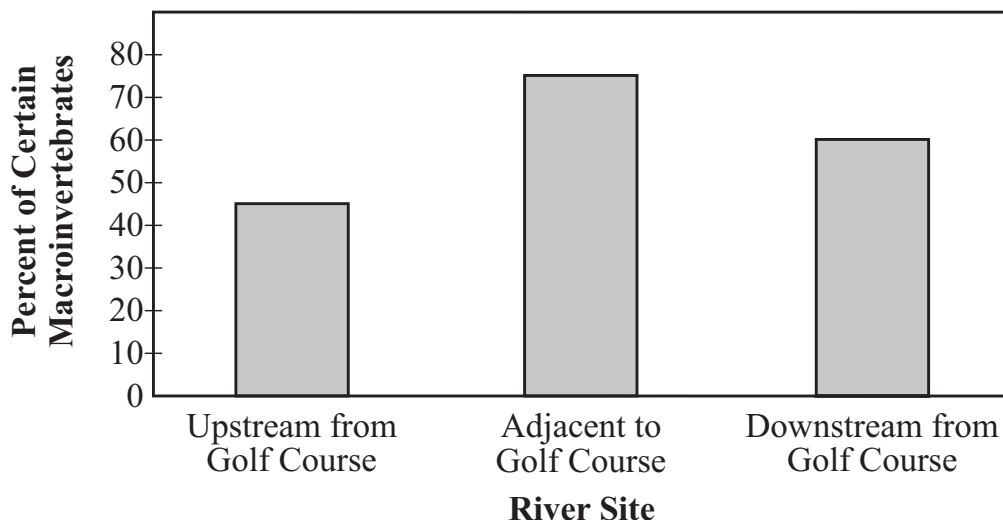
On two days, students collected water samples from three locations along the river: upstream from the golf course, adjacent to the golf course, and downstream from the golf course. At each location, students tested the samples for nitrogen, phosphorus, and dissolved oxygen levels. They also collected data on the percentage of certain macroinvertebrates associated with healthy aquatic systems, such as mayflies, stoneflies, and caddisflies, relative to the percentage of a different fly species.

The table and graph below show the averages for the data collected.

AVERAGE WATER QUALITY MEASUREMENTS AT THREE LOCATIONS

Level Tested	Upstream from Golf Course	Adjacent to Golf Course	Downstream from Golf Course
Total Nitrogen (milligrams/liter)	0.40	0.46	0.44
Total Phosphorus (micrograms/liter)	12.10	16.70	16.90
Dissolved Oxygen (milligrams/liter)	10.00	10.10	11.20

MACROINVERTEBRATES ASSOCIATED WITH HEALTHY AQUATIC ECOSYSTEMS



To answer this question, refer to the “Fertilizer Runoff and Water Quality” passage, table, and graph.

11. Which of the following **best** explains why the amounts of these compounds are monitored in the environment?
- Ⓐ to regulate water flow in the rivers
 - Ⓑ to count the number of macroinvertebrates
 - Ⓒ to determine the amount of fertilizer that is wasted
 - Ⓓ to ensure that acceptable water quality levels are maintained

To answer this question, refer to the “Fertilizer Runoff and Water Quality” passage, table, and graph.

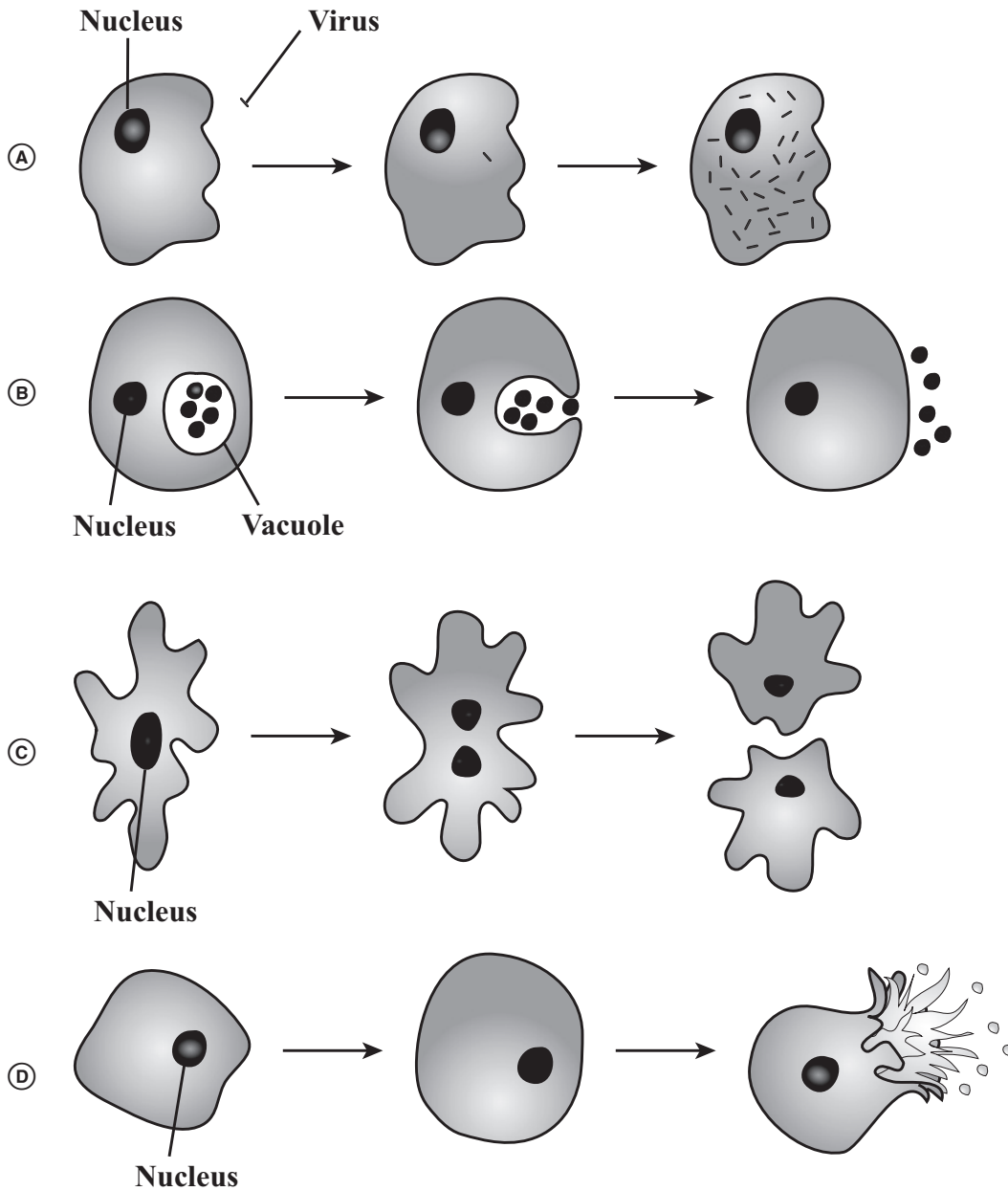
12. If the golf course added fertilizer more often, which of the following predictions about the downstream site would be **most** consistent with the data presented?
- Ⓐ The total nitrogen levels would decrease.
 - Ⓑ The total phosphorus levels would increase.
 - Ⓒ The dissolved oxygen levels would remain the same.
 - Ⓓ The percentage of macroinvertebrates would remain the same.

To answer this question, refer to the “Fertilizer Runoff and Water Quality” passage, table, and graph.

13. Despite the results of this investigation, community members are concerned that the continued use of these compounds will have a negative impact on the river ecosystem. Which of the following actions will **best** provide a long-term solution to sustaining the life in the river?

- Ⓐ Add more fish to the river.
- Ⓑ Decrease the growth of algae in the river.
- Ⓒ Add more plants between the golf course and the river.
- Ⓓ Increase the downward slope from the golf course toward the river.

14. The cell theory is an important concept in biology. Which diagram **best** represents a component of the cell theory?



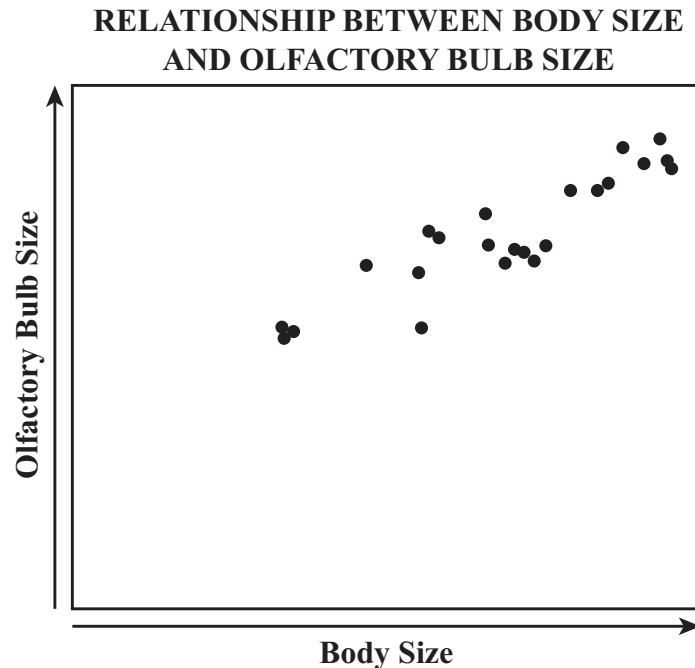
15. The diagram below shows an important cellular process.



What is a role of the cellular process shown?

- Ⓐ to initiate cell growth during earlier mitotic stages
- Ⓑ to transfer phenotypes from parent cell to daughter cells
- Ⓒ to trigger membrane dissolution during later mitotic stages
- Ⓓ to transmit genetic information from parent cell to daughter cell

- 16.** The olfactory bulb is the structure in the brain that is responsible for the sense of smell. Scientists used fossils of extinct dinosaurs and prehistoric birds to estimate the size of their olfactory bulbs and their bodies. The graph below shows their data.



Which of the following is the **best** inference from these data?

- Ⓐ Smaller organisms may have detected different kinds of odors than larger organisms did.
- Ⓑ Larger organisms detected odors with stronger scents compared with other types of organisms.
- Ⓒ The largest of these organisms may have been able to detect more odors than the smallest of these organisms did.
- Ⓓ Organisms of moderate sizes have the least range of ability to detect odors when compared with smaller and larger organisms.

Biology 1

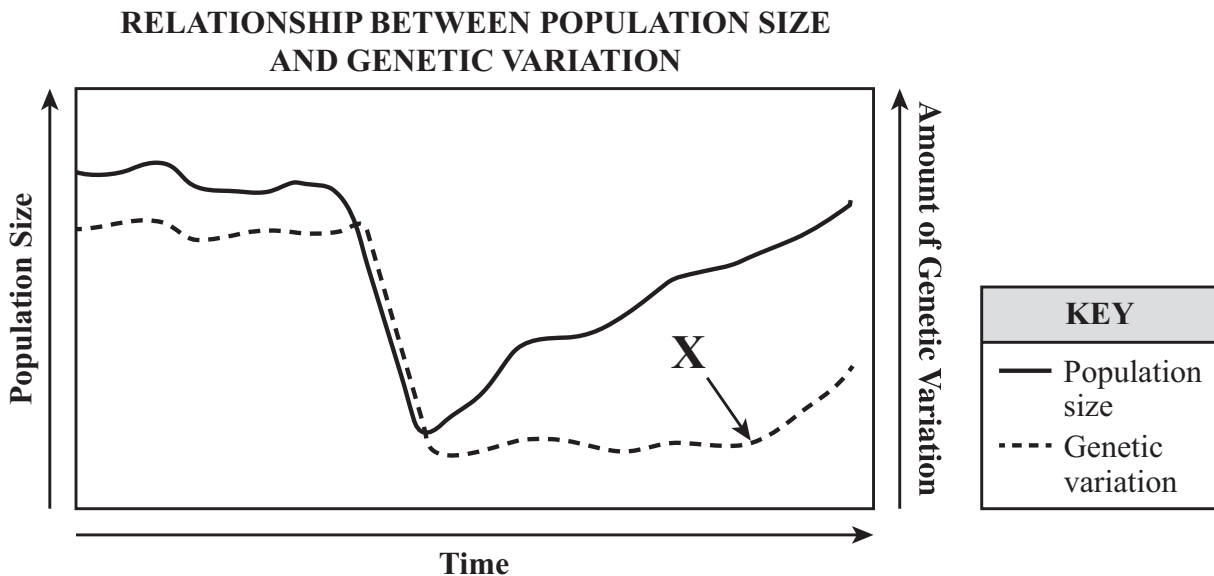
17. Cellular metabolism requires constant transfers of energy throughout the cell. Which of the following statements explains how energy is released for cellular use?

- Ⓐ Cells release the three phosphates attached to ADP one at a time to provide energy.
- Ⓑ The removal of a phosphate from ATP releases energy for metabolic activity.
- Ⓒ Cellular energy is released when additional phosphates are bonded to ATP.
- Ⓓ Energy is released as ADP, and a phosphate is broken down into ATP.

18. Chickens of a certain breed have either black feathers ($C^B C^B$), white feathers ($C^W C^W$), or blue-gray feathers ($C^B C^W$). When two chickens of this breed are crossed, they produce offspring that either have white feathers or blue-gray feathers. Which of the following parental genotypes would produce offspring with these phenotypes?

- Ⓐ $C^B C^B$ and $C^W C^W$
- Ⓑ $C^B C^W$ and $C^W C^W$
- Ⓒ $C^B C^B$ and $C^B C^W$
- Ⓓ $C^B C^W$ and $C^B C^W$

19. When a population dramatically declines in size, the amount of genetic variation in the population also declines. Often, the population size then increases, but the amount of genetic variation does not immediately increase, as shown in the graph below.



Which of the following would **most** contribute to the increase in the amount of genetic variation that begins at point X?

- Ⓐ The amount of gene flow between populations of different organisms decreases.
- Ⓑ The number of chromosomes in gametes of the remaining organisms increases.
- Ⓒ The amount of random mating within the population of organisms decreases.
- Ⓓ The number of DNA mutations that are beneficial to the organisms increases.

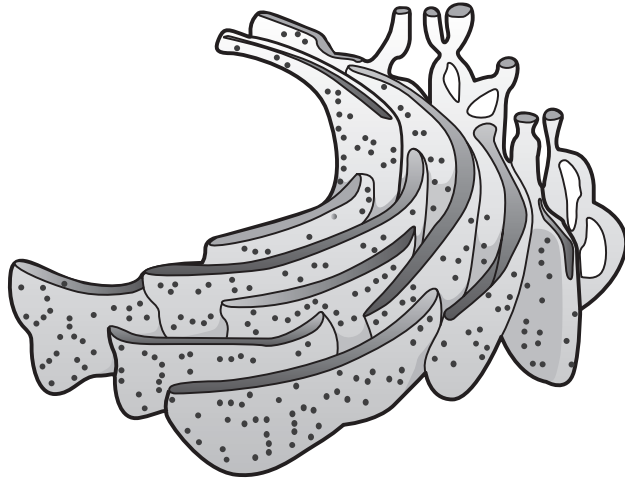
- 20.** Vacuoles are membrane-bound sacs found within the cytoplasm of both plant and animal cells. Which of the following **best** describes how the vacuoles in plant cells differ from the vacuoles in animal cells?
- Ⓐ Vacuoles in plant cells are larger than vacuoles in animal cells because they contain greater amounts of water that help support the cells.
 - Ⓑ Vacuoles in plant cells are larger than vacuoles in animal cells because they contain pigments that are vital to the process of photosynthesis.
 - Ⓒ Vacuoles in plant cells are smaller and more numerous than vacuoles in animal cells because they contain molecules that fight off foreign invaders.
 - Ⓓ Vacuoles in plant cells are smaller and stacked closer together than vacuoles in animal cells because they have proteins that transport materials throughout the cells.

Biology 1

21. The heat of lava from an erupting volcano melts the remaining snow and ice on its slopes. This results in massive flows of mud and water that erode riverbanks and remove many plant species near the riverbanks. Which of the following is the **most likely** immediate consequence of this type of local loss of biodiversity?

- Ⓐ The rate of nutrient recycling would increase along riverbanks.
- Ⓑ Secondary consumers would begin to move into areas near riverbanks.
- Ⓒ Herbivores that feed along riverbanks would need to find food in other places.
- Ⓓ The amount of energy available to consumers along riverbanks would increase.

- 22.** The illustration below shows the structure of the rough endoplasmic reticulum.



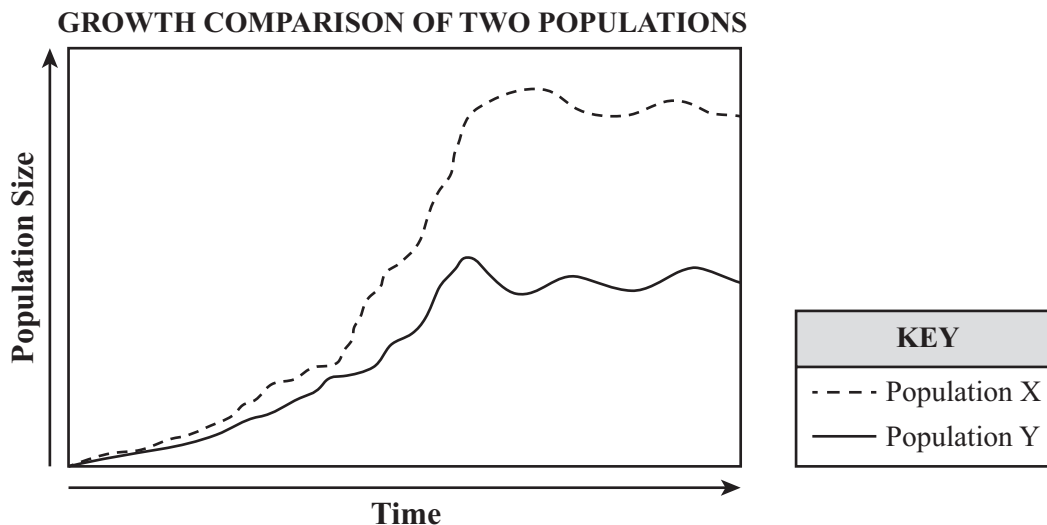
Rough Endoplasmic Reticulum

Which statement describes how the structure of the rough endoplasmic reticulum aids overall cell function?

- Ⓐ The wide layers provide a site for the synthesis of spindle fibers.
- Ⓑ The multiple folds provide ample surface area for protein synthesis.
- Ⓒ The irregular shape allows for the engulfing of waste products for excretion.
- Ⓓ The interconnecting membrane allows for the transport of DNA to other cell organelles.

Biology 1

23. The graph below represents the growth of two different populations of a grazing vertebrate herbivore species. These populations inhabit two different ecosystems that are of similar size.



Which of the following statements **best** relates the carrying capacity of each ecosystem to the long-term growth patterns of these two populations?

- Ⓐ Competition for access to water is less in the ecosystem of population Y, resulting in a lower carrying capacity than the ecosystem of population X.
- Ⓑ The ecosystem of population Y contains fewer predators that consume less prey than the ecosystem of population X, resulting in a lower carrying capacity.
- Ⓒ Additional members of the species are more able to migrate into the ecosystem of population X, resulting in a greater carrying capacity than the ecosystem of population Y.
- Ⓓ The ecosystem of population X receives more rain that produces greater amounts of plant growth than the ecosystem of population Y, resulting in a greater carrying capacity.

24. Animals feed on plants that contain starch, which is broken down into glucose. Which of the following is the primary function glucose serves in an animal's body?

- Ⓐ facilitating cellular transport
- Ⓑ providing a source of energy
- Ⓒ regulating chemical reactions
- Ⓓ transmitting signals between cells

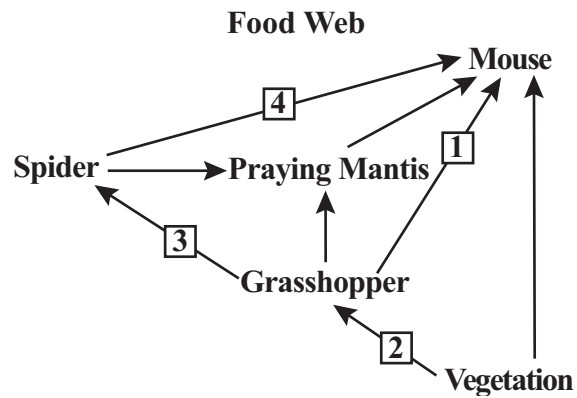
Biology 1

25. Rochelle observes a slide of an onion root tip through a microscope. She wants to identify the number of cells in each phase of mitosis. Which of the following does Rochelle observe in the cells that are in metaphase?

- Ⓐ chromosomes that are at opposite poles of the cell
- Ⓑ chromosomes that are aligned in the middle of the cell
- Ⓒ chromosomes that are surrounded by the nuclear membrane of the cell
- Ⓓ chromosomes that are located between the equator and the poles of the cell

- 26.** Early scientists observed that flies developed from food sources. They formed the idea that organisms can grow from nonliving matter. Later scientists showed that organisms did not grow in sterilized food that was covered, but they did grow if the sterilized food was uncovered. These scientists formed the idea that living organisms can only be reproduced from other living organisms. Which of the following **best** describes why the work of these later scientists meets the criteria to be considered science?
- Ⓐ The later scientists controlled the variables of an experiment instead of relying on observations.
 - Ⓑ The later scientists were trained in how to make an observation instead of how to form a hypothesis.
 - Ⓒ The later scientists based their conclusion on logic and reasoning instead of gathering additional evidence.
 - Ⓓ The later scientists presented a specific set of evidence instead of forming a valid conclusion for a wide range of situations.

27. The food web below shows different energy pathways.



Four of the pathways are labeled. Which label in this food web represents the transfer of the **greatest** amount of the original energy?

- Ⓐ 1
- Ⓑ 2
- Ⓒ 3
- Ⓓ 4

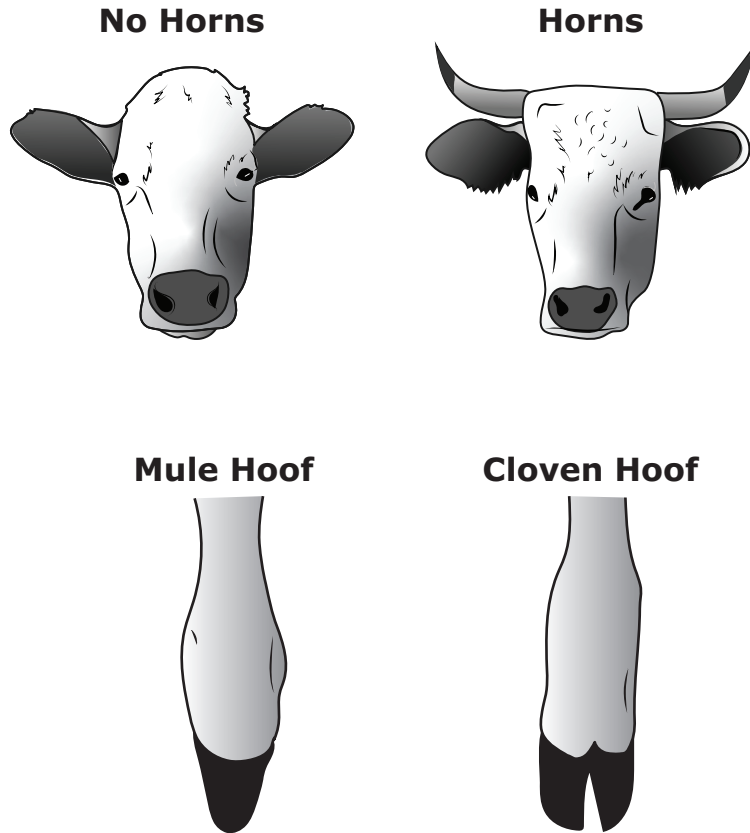
28. Gene mutations may cause changes to phenotypes. Such mutations can be inherited. Which of the following must occur for a gene mutation to cause a phenotypic change that is passed on to the next generation?

- Ⓐ The mutation must be caused by radiation.
- Ⓑ The mutation must affect the DNA of a gamete.
- Ⓒ The mutation must affect the DNA of a somatic cell.
- Ⓓ The mutation must be caused by a random replication error.

29. At specific times in the cell cycle, cells contain chromosomes with sister chromatids. Mistakes sometimes happen during the cell cycle, and the sister chromatids do not separate from each other. Which of the following **best** explains how an improper separation of sister chromatids can lead to phenotypic changes in an organism?

- Ⓐ The chromosomes in daughter cells have unequal numbers of mutations.
- Ⓑ The chromosomes in daughter cells do not contain centromeres.
- Ⓒ Daughter cells cannot use the chromosomes during replication.
- Ⓓ Daughter cells contain different numbers of chromosomes.

- 30.** In some breeds of cattle, the presence of horns is determined by a single gene with two alleles, and the type of hooves is determined by a different gene with two alleles. The illustrations below represent the two phenotypes for each trait. These illustrations are not to scale.



Scientists conducted two crosses between male and female cattle with different phenotypes. The crosses and the approximate percentage of each type of offspring from each cross are listed in the table below.

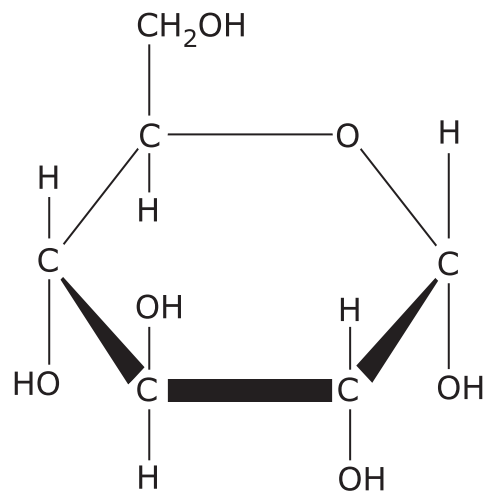
RESULTS OF CATTLE CROSSES

Cross 1		
Male	Female	Offspring
• Heterozygous for no horns • Heterozygous for cloven hooves	• Homozygous for horns • Homozygous for mule hooves	25% no horns and cloven hooves
		25% no horns and mule hooves
		25% horns and cloven hooves
		25% horns and mule hooves
Cross 2		
Male	Female	Offspring
• Homozygous for horns • Homozygous for mule hooves	• Heterozygous for no horns • Heterozygous for cloven hooves	25% no horns and cloven hooves
		25% no horns and mule hooves
		25% horns and cloven hooves
		25% horns and mule hooves

Which of the following **best** describes the inheritance pattern of the traits?

- Ⓐ Having horns and having mule hooves are both autosomal recessive.
- Ⓑ Not having horns and having cloven hooves are both sex-linked dominant.
- Ⓒ Having horns is autosomal recessive, but having mule hooves is sex-linked recessive.
- Ⓓ Not having horns is autosomal dominant, but having cloven hooves is autosomal recessive.

31. As part of an investigation, a student drew the structure shown below.



What type of biological molecule did the student draw?

- Ⓐ a fatty acid
- Ⓑ a nucleotide
- Ⓒ an amino acid
- Ⓓ a monosaccharide

- 32.** Scientists have determined that approximately 20 percent of the blood in the human body flows to muscles when at rest. Which of the following **best** explains how blood flow is expected to change during exercise?
- Ⓐ Exercise increases the heart rate, which causes more blood to flow to the muscles.
 - Ⓑ Exercise increases blood pressure, which decreases the rate of blood flow to the muscles.
 - Ⓒ Exercise causes blood vessels to contract, which speeds up the flow of blood to the muscles.
 - Ⓓ Exercise results in a loss of water, which slows the amount of blood that flows to the muscles.

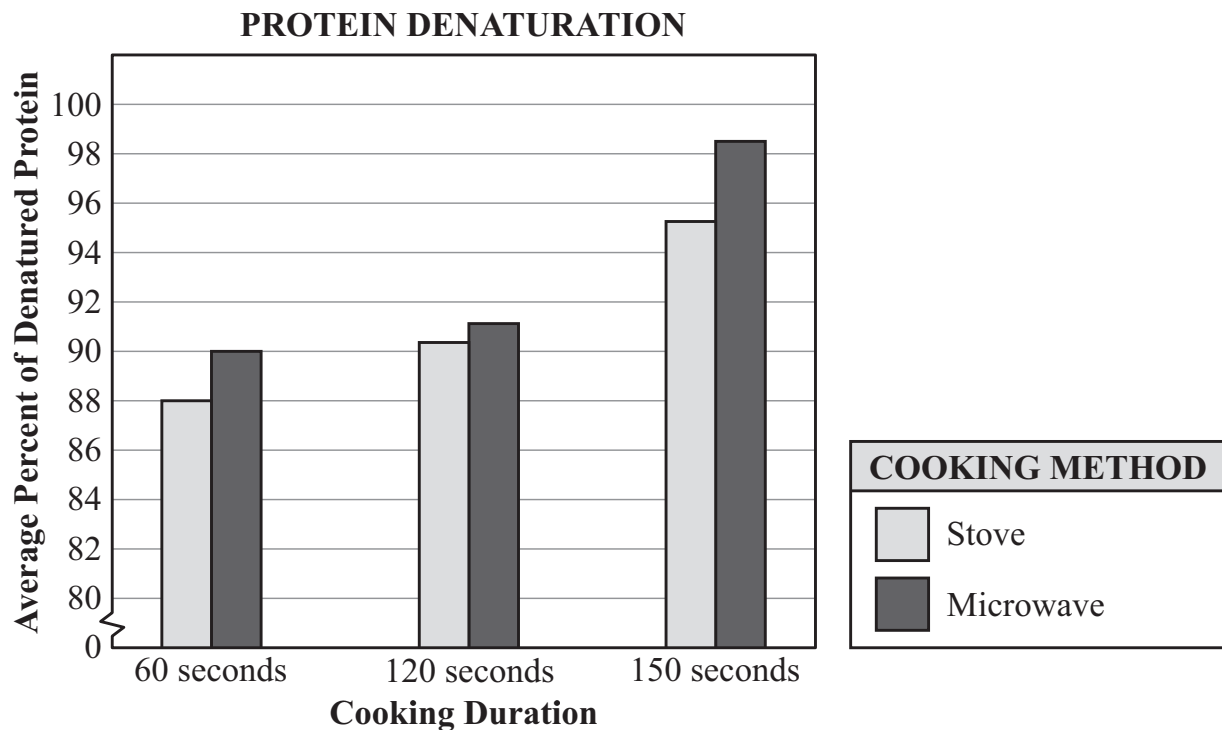
- 33.** A student compared the structures of plant cells and animal cells. After creating a list of similarities and differences between the cells, the student wrote the sentence below.

Animal cells have mitochondria, but plant cells do not.

Which statement **best** evaluates this claim?

- Ⓐ Correct—animal cells have mitochondria, and plant cells do not need to break down glucose.
- Ⓑ Correct—animal cells have mitochondria, and plant cells use chloroplasts to break down glucose.
- Ⓒ Incorrect—plant cells and animal cells have mitochondria because both cell types need to break down glucose.
- Ⓓ Incorrect—plant cells have mitochondria and animal cells do not because animal cells do not need to break down glucose.

34. Cooking eggs denatures the structure of the protein in egg whites. When the protein is denatured, the egg white changes appearance from clear to white. A student investigated how cooking egg whites on a stove and cooking egg whites in a microwave affected the amount of protein that was denatured. The student used an instrument to measure the percent of denatured protein in a sample. Each cooking method was repeated on numerous samples, and the average measurements are shown in the graph below.



Which of the following is the **best** analysis of these data?

- (A) The amount of protein that appears to be denatured in egg whites was less in the microwave than on the stove.
- (B) Similar amounts of protein appear to be denatured in egg whites cooked on the stove for 120 seconds and for 150 seconds.
- (C) Less protein appears to be denatured in the egg whites cooked for 150 seconds than in those cooked for shorter durations of time.
- (D) More protein appears to be denatured in egg whites cooked in the microwave between 120 seconds and 150 seconds than in those cooked on the stove.

- 35.** Horses can have many different shades of coat colors. One of the basic coat colors of horses is red. An additional gene, with a mutation called cream, modifies the red color. The chart below shows the effect that different numbers of the cream mutation have on the coat color of a red horse.

EFFECT OF CREAM MUTATION ON COAT COLOR

Basic Coat Color	Number of Cream Mutation Alleles	Final Coat Color
Red	0	Red
Red	1	Palomino
Red	2	Cremello

If two palomino horses mate, what percentage of offspring are expected to be cremello in color?

- Ⓐ 0%
- Ⓑ 25%
- Ⓒ 50%
- Ⓓ 75%

Biology 1

- 36.** A student uses a classroom microscope to observe the cells of an organism. The student claims that the cells are from a plant. Which of the following student observations provides evidence to support the claim?
- Ⓐ Each of the cells contains cytoplasm.
 - Ⓑ Each of the cells contains chloroplasts.
 - Ⓒ None of the cells contain mitochondria.
 - Ⓓ None of the cells contain a central vacuole.

- 37.** Between 1945 and 1952, the brown tree snake was accidentally introduced to Guam, a small island in the Pacific Ocean. This snake population has increased rapidly since its introduction. With a growing number of these snakes on Guam, many native bird species have become extinct and other native organisms are threatened. Which of the following inferences can **best** be supported by these observations?
- Ⓐ The number of nonnative predators may decrease when competition between predatory species is absent in an ecosystem.
 - Ⓑ Introducing a nonnative species to an isolated island may increase the biodiversity of natural predators found on that island.
 - Ⓒ Introducing a nonnative species into an ecosystem may cause a decrease in biodiversity of other species that are without natural predators.
 - Ⓓ The number of native predators may increase when nonnative species compete with natural predators for resources on an isolated island.

Biology 1

38. Groups of students investigated the effect of detergents on algal growth. Each group used the procedure listed below to compare detergents with and without phosphates.

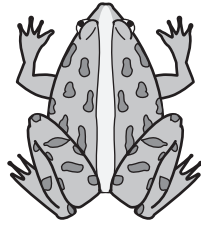
1. Label two 1000-milliliter (mL) beakers 1 and 2, then add 500 mL of pond water that contains algae to each beaker.
2. Use separate, clean pipettes to collect a 1 mL sample from each beaker, then place each sample on a separate slide.
3. Use a compound microscope to observe the algae, then shade in how much of the field of view has algal growth.
4. Add 10 drops of detergent containing phosphate to beaker 1. Mix gently.
5. Add 10 drops of detergent that does not contain phosphate to beaker 2. Mix gently.
6. Place both beakers near a sunny window.
7. Each day for 5 days, collect a 1 mL sample from each beaker, then shade in the field of view to indicate the estimated final algal growth.
8. Compare the results with those of other groups of students.

Which of the following describes the **most** significant flaw in this investigation?

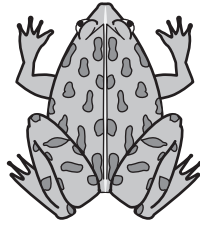
- Ⓐ In the investigation, the beakers were not kept near a sunny window for a long enough time period.
- Ⓑ The investigation does not include a beaker that contains pond water without any detergent.
- Ⓒ The investigation does not include a beaker that contains different types of algae.
- Ⓓ In the investigation, the beakers were not maintained at the same temperature.

- 39.** In the African grass frog, the expression of multiple genes, including a gene called *ASIP*, is responsible for three different spinal stripe phenotypes. The illustration below shows frogs with the three observed phenotypes.

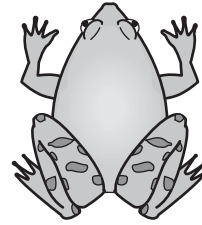
AFRICAN GRASS FROG SPINAL STRIPE PHENOTYPES



Wide Stripe



Thin Stripe



No Stripe

Scientists observed and counted frogs living on land surfaces and frogs living in trees. The scientists claim that the spinal stripe may be selected for when frogs live on land surfaces and selected against when frogs live in trees. How should this claim be evaluated?

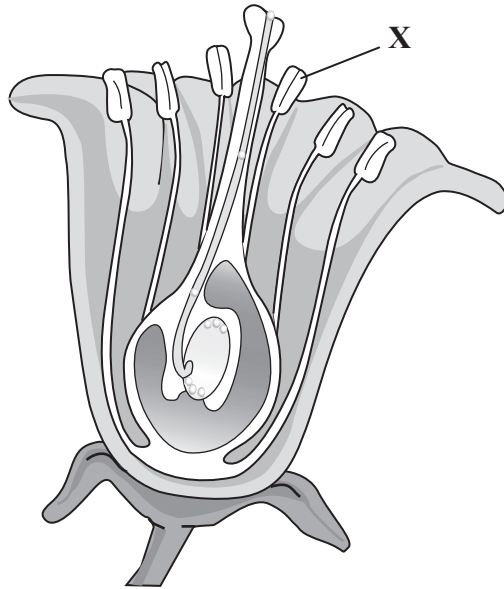
- Ⓐ Scientists should measure the spinal stripe width and calculate the average width for land-dwelling frogs and for tree-dwelling frogs.
- Ⓑ Scientists should confirm that the frogs living on land surfaces and the frogs living in trees express the *ASIP* gene from the same chromosomal location.
- Ⓒ Scientists should determine whether there is a difference in the frequency of expression of the *ASIP* gene between land-dwelling frogs and tree-dwelling frogs.
- Ⓓ Scientists should compare species of frogs living on land surfaces and species of frogs living in trees to determine whether other populations have a spinal stripe.

Biology 1

40. Genetic drift can result in fewer types of phenotypes among a species. Which of the following describes how genetic drift can result in this change?

- Ⓐ It reduces genetic variation.
- Ⓑ It decreases the size of the chromosomes.
- Ⓒ It reduces the number of genes translated on the ribosomes.
- Ⓓ It decreases the number of genes transcribed in the nucleus.

41. Shown below is a cross section of a flower with one part labeled X.



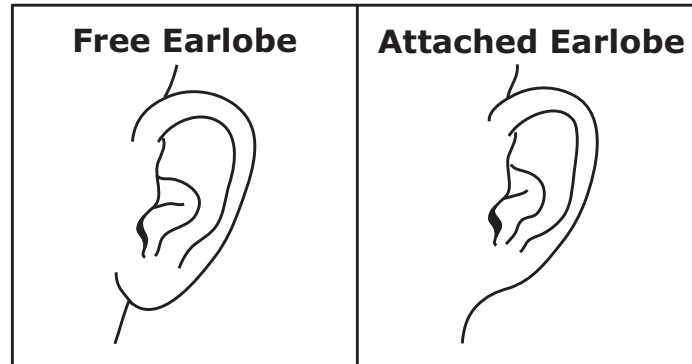
How is the structure of part X related to its function?

- Ⓐ Encasing the egg, it provides protective storage.
- Ⓑ Its green color provides for efficient light absorption.
- Ⓒ Its oval shape attracts birds searching for bugs or seeds.
- Ⓓ Protruding from the surface, it is accessible to pollinators.

Biology 1

- 42.** After fertilization occurs, a human embryo requires multiple months to complete fetal development. Which statement describes an important function of the amniotic fluid during fetal development?
- Ⓐ It supplies oxygen to the fetus.
 - Ⓑ It protects the fetus from injury.
 - Ⓒ It transfers nutrients to the fetus.
 - Ⓓ It separates the fetus from waste.

- 43.** Some people have classified human earlobes as either being free or attached. Two sets of researchers investigated the inheritance of earlobe type by categorizing a large number of offspring from many sets of parents. The earlobe types and the results of the two investigations are shown below.



FREE vs. ATTACHED EARLOBES

Parent 1/Parent 2	Investigation 1			Investigation 2		
	Number of Offspring		Percentage with Free Earlobes	Number of Offspring		Percentage with Free Earlobes
	Free Earlobes	Attached Earlobes		Free Earlobes	Attached Earlobes	
Free/Free	13	1	93	11	21	34
Free/Attached	9	9	50	68	111	38
Attached/Attached	8	34	19	35	90	28

Which of the following is a correct analysis of these data?

- Ⓐ The percentage of offspring with free earlobes is similar for all sets of parents in the investigations.
- Ⓑ A comparison of Investigation 1 and Investigation 2 shows that offspring of two parents with attached earlobes are most likely to have free earlobes.
- Ⓒ A comparison of the investigations shows that the largest difference in the percentage of offspring with free earlobes occurs when both parents have free earlobes.
- Ⓓ The percentage difference between offspring having free earlobes in the investigations is greater when the parents have different types of earlobes than when they have the same type of earlobes.

- 44.** Some viruses enter the human body and attack the immune system. The immune system is then unable to remove these viruses, so they remain active in the body for a long time. Scientists have developed treatments using different types of medications to remove a particular virus from the body. The chart below describes the actions of five types of these medications.

ACTIONS OF MEDICATIONS

Type	Action
1	Turns off a protein that the virus needs to make copies of itself
2	Blocks the virus from entering some types of human immune cells
3	Interrupts the function of a protein that the virus needs to make copies of itself
4	Provides nonfunctional pieces of the building blocks that the virus uses to make copies of itself
5	Disables a protein that the virus uses to insert genetic material into some types of human immune cells

Which statement **best** explains why patients who carry this kind of virus are prescribed a long-term treatment of more than one type of the listed medications?

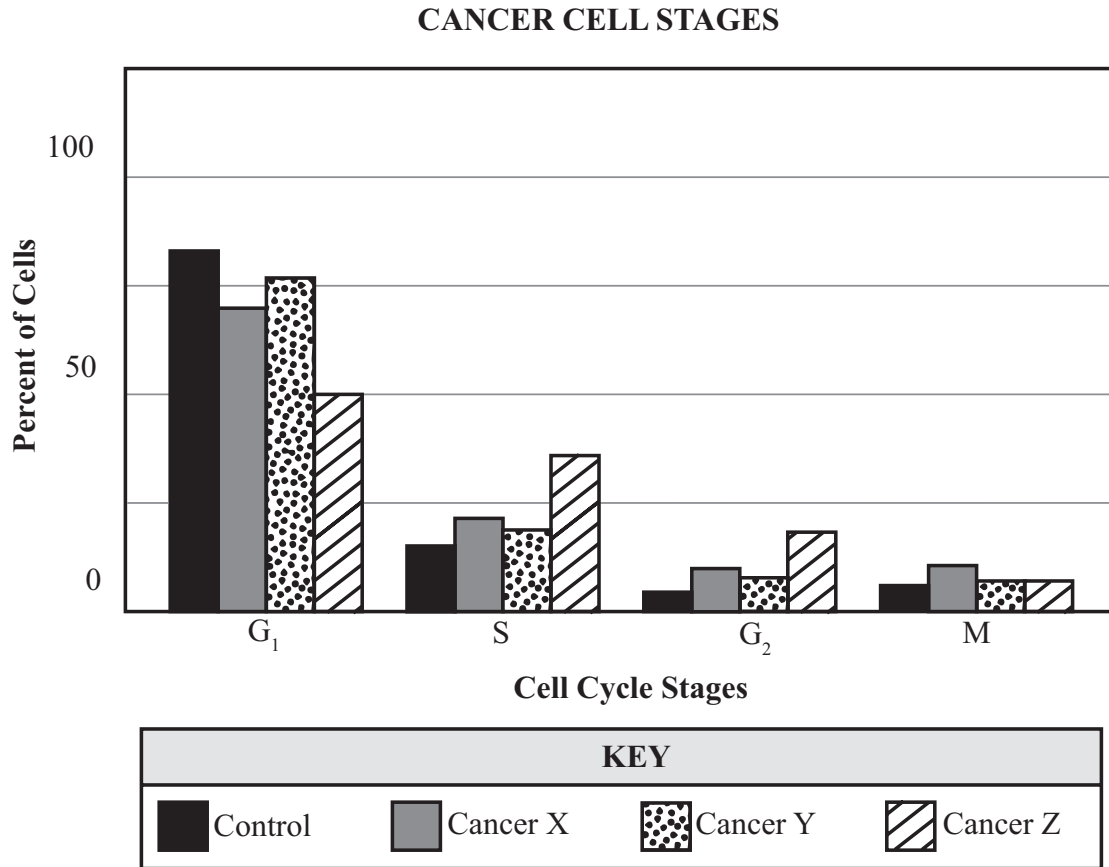
- Ⓐ Using more than one type of these medications causes the virus to rearrange its genetic material before it is replicated.
- Ⓑ Using more than one type of these medications prevents the virus from copying its genetic material at a faster rate than normal.
- Ⓒ Using more than one type of these medications helps the virus enter the parts of the body that are resistant to one type of medication.
- Ⓓ Using more than one type of these medications prevents the formation of versions of the virus that are resistant to any specific type of medication.



Biology 1 EOC Assessment

Test Release: 2026

1. Cancer research involves the investigation of many cell factors. One factor is the stage of the cell cycle the cancer cells are in. A scientist conducted an investigation using various forms of cancer cells to compare the percentage of cells that are in each stage of the cell cycle. The graph below shows a summary of the data from one study.



Which analysis of these data can be made?

- Ⓐ The cancer cells spend less time in the G₂ stage than the control cells do.
- Ⓑ There is a greater percentage of Cancer Y cells than Cancer X cells in the M stage.
- Ⓒ There is a lower percentage of any type of cancer cells than the control cells in the G₁ stage.
- Ⓓ The control and Cancer Y cells spend more time in the S stage than Cancer X and Cancer Z cells do.

Biology 1 Test Release Support Document

Answer Key: C

Percentage of Students Answering Correctly: 75%

Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.N.1.1

Benchmark Description: Define a problem based on a specific body of knowledge, for example: biology, chemistry, physics, and earth/space science, and do the following:

1. pose questions about the natural world,
2. conduct systematic observations,
3. examine books and other sources of information to see what is already known,
4. review what is known in light of empirical evidence,
5. plan investigations,
6. use tools to gather, analyze, and interpret data (this includes the use of measurement in metric and other systems, and also the generation and interpretation of graphical representations of data, including data tables and graphs),
7. pose answers, explanations, or descriptions of events,
8. generate explanations that explicate or describe natural phenomena (inferences),
9. use appropriate evidence and reasoning to justify these explanations to others,
10. communicate results of scientific investigations, and
11. evaluate the merits of the explanations produced by others.

2. The process of meiosis is one way the cells of organisms prepare to divide. Which of the following is a characteristic of meiosis that results in genetic variation?
- Ⓐ division of somatic cells
 - Ⓑ formation of spindle fibers
 - Ⓒ replication of genetic material
 - Ⓓ crossing over of chromosomes

Answer Key: D

Percentage of Students Answering Correctly: 56%

Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.L.16.16

Benchmark Description: Describe the process of meiosis, including independent assortment and crossing over. Explain how reduction division results in the formation of haploid gametes or spores.

Biology 1 Test Release Support Document

3. In 1995, scientists transferred a gene from wild rice into a type of cultivated rice. The expression of the transferred gene resulted in the cultivated rice being resistant to a specific bacterium that causes a deadly disease in rice plants. Which of the following **best** describes a benefit to society of this example of biotechnology?
- Ⓐ a reduction in the amount of nutrients removed from rice fields
 - Ⓑ a reduction in the need to apply fertilizers to rice fields
 - Ⓒ an increase in tolerance of rice for drought conditions
 - Ⓓ an increase in the production of rice for food

Answer Key: D

Percentage of Students Answering Correctly: 44%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.L.16.10

Benchmark Description: Evaluate the impact of biotechnology on the individual, society, and the environment, including medical and ethical issues.

4. Three varieties of mangroves are found along the coastal regions of Florida. The red and the white mangroves inhabit the eastern and western coastal areas as far north as 29° latitude. The black mangrove overlaps with the other two varieties and grows even farther north, up to 30° latitude. Which of the following factors **most likely** contributes to the more northern aquatic range of the black mangrove?
- Ⓐ The black mangrove survives in colder water temperatures.
 - Ⓑ The black mangrove prefers the shallower coastal waters in the north.
 - Ⓒ The Gulf Stream provides nutrient-rich waters for the black mangrove.
 - Ⓓ The more northern waters provide the black mangrove with proper salinity levels.

Answer Key: A

Percentage of Students Answering Correctly: 44%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.L.17.2

Benchmark Description: Explain the general distribution of life in aquatic systems as a function of chemistry, geography, light, depth, salinity, and temperature.

Biology 1 Test Release Support Document

5. A patient visits a physician because of a sore throat. The physician diagnoses the patient with an infection and determines that antibiotics will not be effective. Which of these is the **most likely** cause of this infection?
- Ⓐ bacteria
 - Ⓑ fungi
 - Ⓒ protozoans
 - Ⓓ viruses

Answer Key: D

Percentage of Students Answering Correctly: 51%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.L.14.52

Benchmark Description: Explain the basic functions of the human immune system, including specific and nonspecific immune response, vaccines, and antibiotics.

6. In December of 1952, the amount of smoke and sulfur dioxide in the air of London, England, dramatically increased. A thick layer of smog covered the city as a result of a lack of wind and of an increase in burning coal to provide heat. The table below shows some of the data collected during the first half of the month.

AIR POLLUTANT LEVELS AND NUMBER OF DEATHS

Date (December 1952)	Sulfur Dioxide Levels (parts per billion)	Smoke Levels (micrograms/cubic meter)	Deaths per Day
1	98	257	257
3	148	445	327
5	375	1096	410
7	655	1595	905
9	452	1212	800
11	127	305	545
13	130	394	508
15	135	352	454

Which statement **best** describes the potential long-term effects on people who live in locations where these types of environmental conditions occur consistently throughout the year?

- Ⓐ Sulfur dioxide and smoke levels that rise quickly result in more respiratory illnesses than if the levels rise slowly.
- Ⓑ Repeated exposure to greater levels of smoke and sulfur dioxide may lead to an increased risk of respiratory problems.
- Ⓒ The elevation of sulfur dioxide levels occurs when smoke levels are also high, leading to illnesses in younger individuals.
- Ⓓ The most harmful level of pollutants affecting human health occurs when the amount of smoke and sulfur dioxide reach the same level.

Biology 1 Test Release Support Document

Answer Key: B

Percentage of Students Answering Correctly: 65%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.L.14.6

Benchmark Description: Explain the significance of genetic factors, environmental factors, and pathogenic agents to health from the perspectives of both individual and public health.

7. In the 1830s, Matthias Schleiden examined the structure of plant cells and Theodor Schwann examined the structure of animal cells. After discussing their findings, they proposed an early version of the cell theory. Which of the following statements **best** describes why developing the cell theory was a scientific process?
- Ⓐ They created a hypothesis that was believed by other scientists.
 - Ⓑ They provided procedures that were approved by other scientists.
 - Ⓒ They made observations that could be repeated by other scientists.
 - Ⓓ They formed a conclusion that was immediately accepted by other scientists.

Answer Key: C

Percentage of Students Answering Correctly: 65%

Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.N.2.1

Benchmark Description: Identify what is science, what clearly is not science, and what superficially resembles science (but fails to meet the criteria for science).

Biology 1 Test Release Support Document

8. Students are asked to classify an unknown organism into the appropriate kingdom. The organism is multicellular with eukaryotic cells arranged into tissues. The cells of the organism do not have cell walls, and the organism is heterotrophic. In which of the following kingdoms should this organism **most likely** be placed?

- ☒ (A) Animalia
- ☐ (B) Fungi
- ☐ (C) Plantae
- ☐ (D) Protista

Answer Key: A

Percentage of Students Answering Correctly: 55%

Reporting Category: Classification, Heredity, and Evolution

Benchmark: SC.912.L.15.6

Benchmark Description: Discuss distinguishing characteristics of the domains and kingdoms of living organisms.

9. Students are investigating the types of fish that live at different ocean depths. Which of the following describes characteristics of a fish that is **best** suited to live in deep ocean environments?
- Ⓐ a fish that is blue on the top and white on the bottom to reduce predation from above and below
 - Ⓑ a dark-colored fish that is adapted to cold water and uses a chemical light source to attract food
 - Ⓒ a blue-colored fish that feeds heavily on the types of algae that could damage a coral-reef system
 - Ⓓ a fish that relies on low-light conditions, on water with a high oxygen level, and on the use of a gas-filled chamber

Answer Key: B

Percentage of Students Answering Correctly: 68%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.L.17.2

Benchmark Description: Explain the general distribution of life in aquatic systems as a function of chemistry, geography, light, depth, salinity, and temperature.

10. A student read an article about the positive effects of extended exercise on the cardiovascular system. The student designed the investigation described below to test the claim that exercise decreased a person's blood pressure.

1. Record the initial blood pressure of 12 participants at noon on day 1 of the investigation.
2. On day 1, each participant exercises for 30 minutes by either walking on a treadmill or riding an exercise bicycle.
3. Each participant alternates the type of exercise performed for 30 minutes for six consecutive days, followed by one day of rest, for 12 weeks.
4. At noon on the seventh day of each week, record the blood pressure of each participant.
5. At the end of 12 weeks, calculate the change in blood pressure for each participant and calculate the average change in blood pressure for the group.

Which of the following would **most** improve the design of the student's investigation?

- Ⓐ Include participants that do not exercise for comparison.
- Ⓑ Have each participant conduct a different type of exercise.
- Ⓒ Allow some participants to exercise for longer periods of time.
- Ⓓ Measure blood pressure of the participants during exercise each day.

Answer Key: A

Percentage of Students Answering Correctly: 49%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.N.1.1

Benchmark Description: Define a problem based on a specific body of knowledge, for example: biology, chemistry, physics, and earth/space science, and do the following:

1. pose questions about the natural world,
2. conduct systematic observations,
3. examine books and other sources of information to see what is already known,
4. review what is known in light of empirical evidence,
5. plan investigations,
6. use tools to gather, analyze, and interpret data (this includes the use of measurement in metric and other systems, and also the generation and interpretation of graphical representations of data, including data tables and graphs),
7. pose answers, explanations, or descriptions of events,
8. generate explanations that explicate or describe natural phenomena (inferences),
9. use appropriate evidence and reasoning to justify these explanations to others,
10. communicate results of scientific investigations, and
11. evaluate the merits of the explanations produced by others.

Fertilizer Runoff and Water Quality

Students in Ms. Macedo's class know that fertilizers add nitrogen-, phosphorus-, and potassium-containing compounds to the soil. They decided to conduct an investigation to determine the effects of fertilizer runoff on the water quality of a river that runs through a wooded area near a golf course.

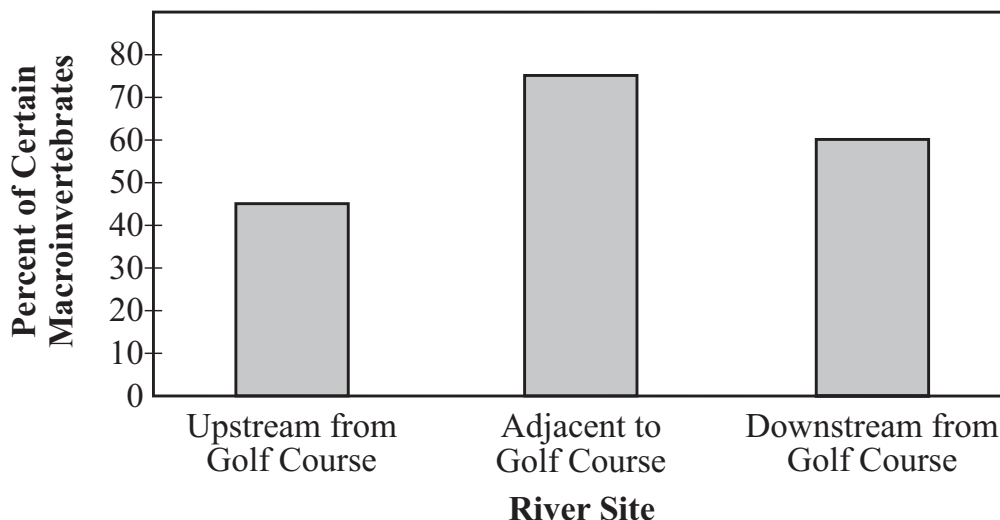
On two days, students collected water samples from three locations along the river: upstream from the golf course, adjacent to the golf course, and downstream from the golf course. At each location, students tested the samples for nitrogen, phosphorus, and dissolved oxygen levels. They also collected data on the percentage of certain macroinvertebrates associated with healthy aquatic systems, such as mayflies, stoneflies, and caddisflies, relative to the percentage of a different fly species.

The table and graph below show the averages for the data collected.

AVERAGE WATER QUALITY MEASUREMENTS AT THREE LOCATIONS

Level Tested	Upstream from Golf Course	Adjacent to Golf Course	Downstream from Golf Course
Total Nitrogen (milligrams/liter)	0.40	0.46	0.44
Total Phosphorus (micrograms/liter)	12.10	16.70	16.90
Dissolved Oxygen (milligrams/liter)	10.00	10.10	11.20

MACROINVERTEBRATES ASSOCIATED WITH HEALTHY AQUATIC ECOSYSTEMS



To answer this question, refer to the “Fertilizer Runoff and Water Quality” passage, table, and graph.

- 11.** Which of the following **best** explains why the amounts of these compounds are monitored in the environment?
- Ⓐ to regulate water flow in the rivers
 - Ⓑ to count the number of macroinvertebrates
 - Ⓒ to determine the amount of fertilizer that is wasted
 - Ⓓ to ensure that acceptable water quality levels are maintained

Answer Key: D

Percentage of Students Answering Correctly: 71%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.L.17.13

Benchmark Description: Discuss the need for adequate monitoring of environmental parameters when making policy decisions.

Biology 1 Test Release Support Document

To answer this question, refer to the “Fertilizer Runoff and Water Quality” passage, table, and graph.

12. If the golf course added fertilizer more often, which of the following predictions about the downstream site would be **most** consistent with the data presented?

- Ⓐ The total nitrogen levels would decrease.
- Ⓑ The total phosphorus levels would increase.
- Ⓒ The dissolved oxygen levels would remain the same.
- Ⓓ The percentage of macroinvertebrates would remain the same.

Answer Key: B

Percentage of Students Answering Correctly: 73%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.N.1.1

Benchmark Description: Define a problem based on a specific body of knowledge, for example: biology, chemistry, physics, and earth/space science, and do the following:

1. pose questions about the natural world,
2. conduct systematic observations,
3. examine books and other sources of information to see what is already known,
4. review what is known in light of empirical evidence,
5. plan investigations,
6. use tools to gather, analyze, and interpret data (this includes the use of measurement in metric and other systems, and also the generation and interpretation of graphical representations of data, including data tables and graphs),
7. pose answers, explanations, or descriptions of events,
8. generate explanations that explicate or describe natural phenomena (inferences),
9. use appropriate evidence and reasoning to justify these explanations to others,
10. communicate results of scientific investigations, and
11. evaluate the merits of the explanations produced by others.

To answer this question, refer to the “Fertilizer Runoff and Water Quality” passage, table, and graph.

13. Despite the results of this investigation, community members are concerned that the continued use of these compounds will have a negative impact on the river ecosystem. Which of the following actions will **best** provide a long-term solution to sustaining the life in the river?

- Ⓐ Add more fish to the river.
- Ⓑ Decrease the growth of algae in the river.
- Ⓒ Add more plants between the golf course and the river.
- Ⓓ Increase the downward slope from the golf course toward the river.

Answer Key: C

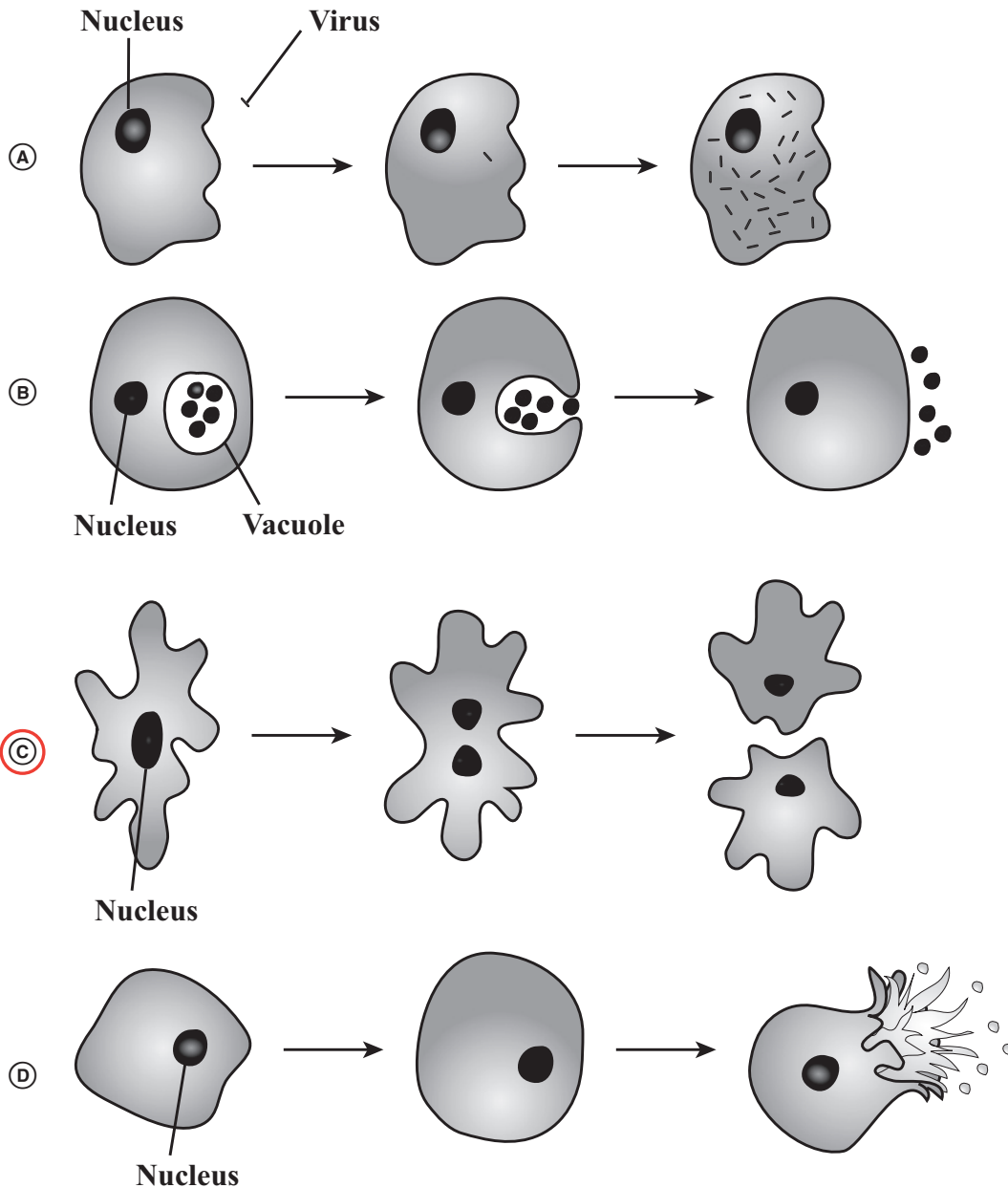
Percentage of Students Answering Correctly: 52%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.L.17.20

Benchmark Description: Predict the impact of individuals on environmental systems and examine how human lifestyles affect sustainability.

14. The cell theory is an important concept in biology. Which diagram **best** represents a component of the cell theory?



Answer Key: C

Percentage of Students Answering Correctly: 64%

Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.L.14.1

Benchmark Description: Describe the scientific theory of cells (cell theory) and relate the history of its discovery to the process of science.

15. The diagram below shows an important cellular process.



What is a role of the cellular process shown?

- Ⓐ to initiate cell growth during earlier mitotic stages
- Ⓑ to transfer phenotypes from parent cell to daughter cells
- Ⓒ to trigger membrane dissolution during later mitotic stages
- Ⓓ to transmit genetic information from parent cell to daughter cell

Answer Key: D

Percentage of Students Answering Correctly: 66%

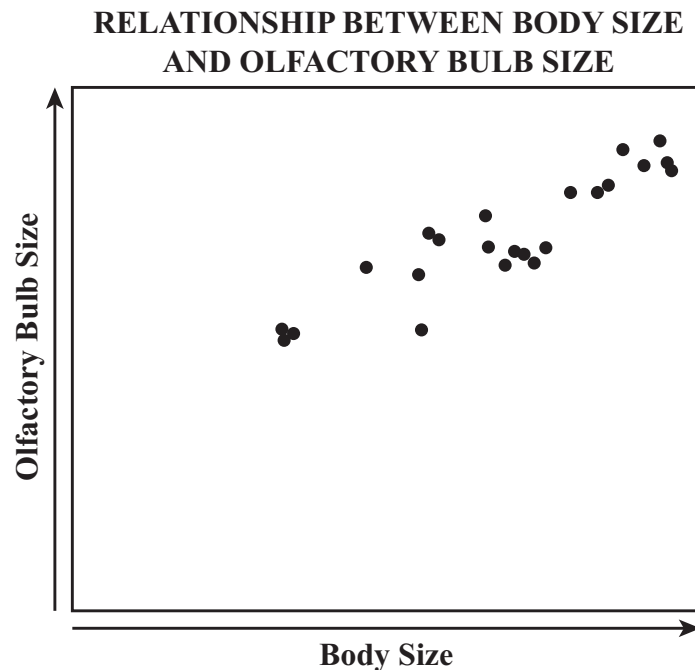
Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.L.16.3

Benchmark Description: Describe the basic process of DNA replication and how it relates to the transmission and conservation of the genetic information.

Biology 1 Test Release Support Document

- 16.** The olfactory bulb is the structure in the brain that is responsible for the sense of smell. Scientists used fossils of extinct dinosaurs and prehistoric birds to estimate the size of their olfactory bulbs and their bodies. The graph below shows their data.



Which of the following is the **best** inference from these data?

- Ⓐ Smaller organisms may have detected different kinds of odors than larger organisms did.
- Ⓑ Larger organisms detected odors with stronger scents compared with other types of organisms.
- Ⓒ The largest of these organisms may have been able to detect more odors than the smallest of these organisms did.
- Ⓓ Organisms of moderate sizes have the least range of ability to detect odors when compared with smaller and larger organisms.

Answer Key: C

Percentage of Students Answering Correctly: 60%

Reporting Category: Classification, Heredity, and Evolution

Benchmark: SC.912.N.1.6

Benchmark Description: Describe how scientific inferences are drawn from scientific observations and provide examples from content being studied.

17. Cellular metabolism requires constant transfers of energy throughout the cell. Which of the following statements explains how energy is released for cellular use?

- Ⓐ Cells release the three phosphates attached to ADP one at a time to provide energy.
- Ⓑ The removal of a phosphate from ATP releases energy for metabolic activity.
- Ⓒ Cellular energy is released when additional phosphates are bonded to ATP.
- Ⓓ Energy is released as ADP, and a phosphate is broken down into ATP.

Answer Key: B

Percentage of Students Answering Correctly: 36%

Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.L.18.10

Benchmark Description: Connect the role of adenosine triphosphate (ATP) to energy transfers within a cell.

Biology 1 Test Release Support Document

18. Chickens of a certain breed have either black feathers ($C^B C^B$), white feathers ($C^W C^W$), or blue-gray feathers ($C^B C^W$). When two chickens of this breed are crossed, they produce offspring that either have white feathers or blue-gray feathers. Which of the following parental genotypes would produce offspring with these phenotypes?

- Ⓐ $C^B C^B$ and $C^W C^W$
- Ⓑ $C^B C^W$ and $C^W C^W$
- Ⓒ $C^B C^B$ and $C^B C^W$
- Ⓓ $C^B C^W$ and $C^B C^W$

Answer Key: B

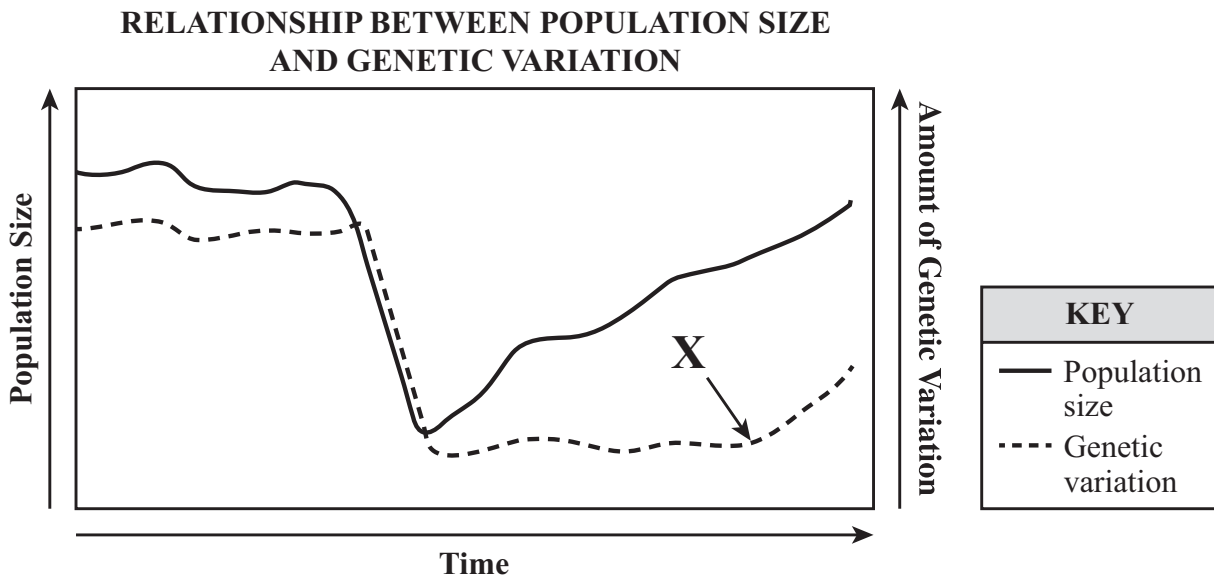
Percentage of Students Answering Correctly: 64%

Reporting Category: Classification, Heredity, and Evolution

Benchmark: SC.912.L.16.1

Benchmark Description: Use Mendel's laws of segregation and independent assortment to analyze patterns of inheritance.

19. When a population dramatically declines in size, the amount of genetic variation in the population also declines. Often, the population size then increases, but the amount of genetic variation does not immediately increase, as shown in the graph below.



Which of the following would **most** contribute to the increase in the amount of genetic variation that begins at point X?

- Ⓐ The amount of gene flow between populations of different organisms decreases.
- Ⓑ The number of chromosomes in gametes of the remaining organisms increases.
- Ⓒ The amount of random mating within the population of organisms decreases.
- Ⓓ The number of DNA mutations that are beneficial to the organisms increases.

Answer Key: D

Percentage of Students Answering Correctly: 51%

Reporting Category: Classification, Heredity, and Evolution

Benchmark: SC.912.L.15.15

Benchmark Description: Describe how mutation and genetic recombination increase genetic variation.

Biology 1 Test Release Support Document

- 20.** Vacuoles are membrane-bound sacs found within the cytoplasm of both plant and animal cells. Which of the following **best** describes how the vacuoles in plant cells differ from the vacuoles in animal cells?
- Ⓐ Vacuoles in plant cells are larger than vacuoles in animal cells because they contain greater amounts of water that help support the cells.
 - Ⓑ Vacuoles in plant cells are larger than vacuoles in animal cells because they contain pigments that are vital to the process of photosynthesis.
 - Ⓒ Vacuoles in plant cells are smaller and more numerous than vacuoles in animal cells because they contain molecules that fight off foreign invaders.
 - Ⓓ Vacuoles in plant cells are smaller and stacked closer together than vacuoles in animal cells because they have proteins that transport materials throughout the cells.

Answer Key: A

Percentage of Students Answering Correctly: 49%

Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.L.14.3

Benchmark Description: Compare and contrast the general structures of plant and animal cells. Compare and contrast the general structures of prokaryotic and eukaryotic cells.

21. The heat of lava from an erupting volcano melts the remaining snow and ice on its slopes. This results in massive flows of mud and water that erode riverbanks and remove many plant species near the riverbanks. Which of the following is the **most likely** immediate consequence of this type of local loss of biodiversity?

- Ⓐ The rate of nutrient recycling would increase along riverbanks.
- Ⓑ Secondary consumers would begin to move into areas near riverbanks.
- Ⓒ Herbivores that feed along riverbanks would need to find food in other places.
- Ⓓ The amount of energy available to consumers along riverbanks would increase.

Answer Key: C

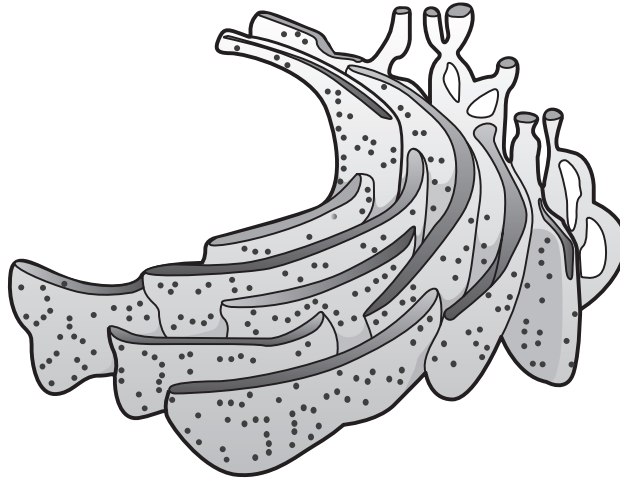
Percentage of Students Answering Correctly: 77%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.L.17.8

Benchmark Description: Recognize the consequences of the losses of biodiversity due to catastrophic events, climate changes, human activity, and the introduction of invasive, non-native species.

22. The illustration below shows the structure of the rough endoplasmic reticulum.



Rough Endoplasmic Reticulum

Which statement describes how the structure of the rough endoplasmic reticulum aids overall cell function?

- Ⓐ The wide layers provide a site for the synthesis of spindle fibers.
- Ⓑ The multiple folds provide ample surface area for protein synthesis.
- Ⓒ The irregular shape allows for the engulfing of waste products for excretion.
- Ⓓ The interconnecting membrane allows for the transport of DNA to other cell organelles.

Answer Key: B

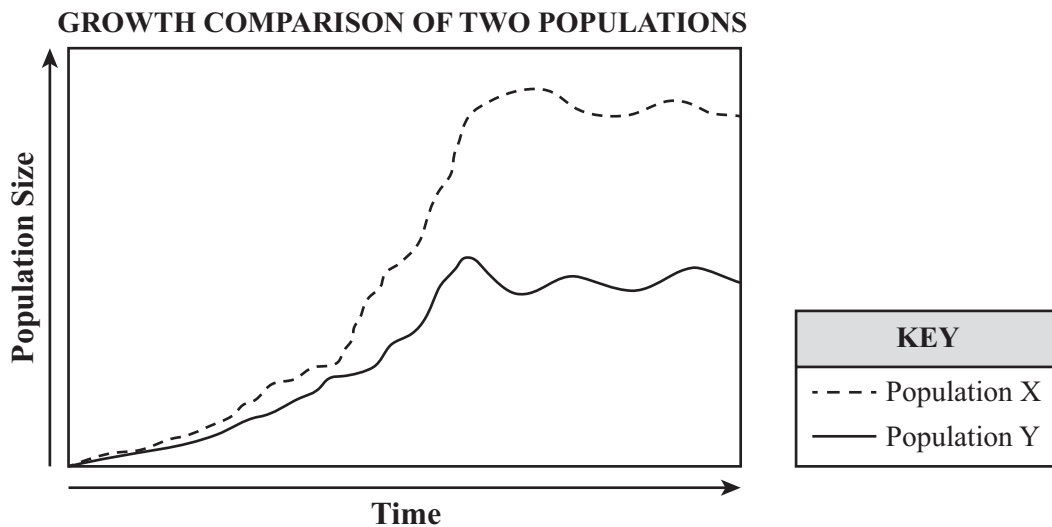
Percentage of Students Answering Correctly: 42%

Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.L.14.2

Benchmark Description: Relate structure to function for the components of plant and animal cells. Explain the role of cell membranes as a highly selective barrier (passive and active transport).

23. The graph below represents the growth of two different populations of a grazing vertebrate herbivore species. These populations inhabit two different ecosystems that are of similar size.



Which of the following statements **best** relates the carrying capacity of each ecosystem to the long-term growth patterns of these two populations?

- Ⓐ Competition for access to water is less in the ecosystem of population Y, resulting in a lower carrying capacity than the ecosystem of population X.
- Ⓑ The ecosystem of population Y contains fewer predators that consume less prey than the ecosystem of population X, resulting in a lower carrying capacity.
- Ⓒ Additional members of the species are more able to migrate into the ecosystem of population X, resulting in a greater carrying capacity than the ecosystem of population Y.
- Ⓓ The ecosystem of population X receives more rain that produces greater amounts of plant growth than the ecosystem of population Y, resulting in a greater carrying capacity.

Biology 1 Test Release Support Document

Answer Key: D

Percentage of Students Answering Correctly: 49%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.L.17.5

Benchmark Description: Analyze how population size is determined by births, deaths, immigration, emigration, and limiting factors (biotic and abiotic) that determine carrying capacity.

24. Animals feed on plants that contain starch, which is broken down into glucose. Which of the following is the primary function glucose serves in an animal's body?

- Ⓐ facilitating cellular transport
- Ⓑ providing a source of energy
- Ⓒ regulating chemical reactions
- Ⓓ transmitting signals between cells

Answer Key: B

Percentage of Students Answering Correctly: 82%

Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.L.18.1

Benchmark Description: Describe the basic molecular structures and primary functions of the four major categories of biological macromolecules.

Biology 1 Test Release Support Document

25. Rochelle observes a slide of an onion root tip through a microscope. She wants to identify the number of cells in each phase of mitosis. Which of the following does Rochelle observe in the cells that are in metaphase?

- Ⓐ chromosomes that are at opposite poles of the cell
- Ⓑ chromosomes that are aligned in the middle of the cell
- Ⓒ chromosomes that are surrounded by the nuclear membrane of the cell
- Ⓓ chromosomes that are located between the equator and the poles of the cell

Answer Key: B

Percentage of Students Answering Correctly: 55%

Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.L.16.14

Benchmark Description: Describe the cell cycle, including the process of mitosis. Explain the role of mitosis in the formation of new cells and its importance in maintaining chromosome number during asexual reproduction.

26. Early scientists observed that flies developed from food sources. They formed the idea that organisms can grow from nonliving matter. Later scientists showed that organisms did not grow in sterilized food that was covered, but they did grow if the sterilized food was uncovered. These scientists formed the idea that living organisms can only be reproduced from other living organisms. Which of the following **best** describes why the work of these later scientists meets the criteria to be considered science?

- Ⓐ The later scientists controlled the variables of an experiment instead of relying on observations.
- Ⓑ The later scientists were trained in how to make an observation instead of how to form a hypothesis.
- Ⓒ The later scientists based their conclusion on logic and reasoning instead of gathering additional evidence.
- Ⓓ The later scientists presented a specific set of evidence instead of forming a valid conclusion for a wide range of situations.

Answer Key: A

Percentage of Students Answering Correctly: 42%

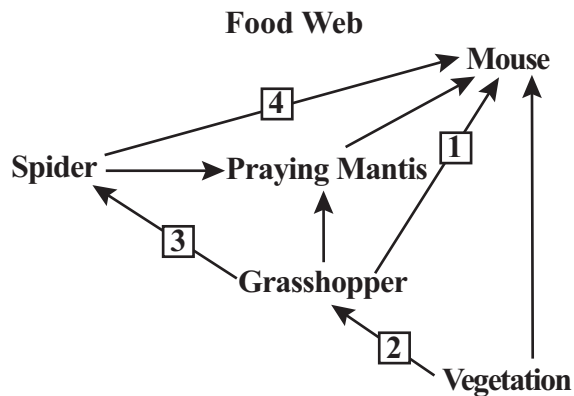
Reporting Category: Classification, Heredity, and Evolution

Benchmark: SC.912.N.2.1

Benchmark Description: Identify what is science, what clearly is not science, and what superficially resembles science (but fails to meet the criteria for science).

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27. The food web below shows different energy pathways.



Four of the pathways are labeled. Which label in this food web represents the transfer of the **greatest** amount of the original energy?

- Ⓐ 1
- Ⓑ 2
- Ⓒ 3
- Ⓓ 4

Answer Key: B

Percentage of Students Answering Correctly: 65%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.L.17.9

Benchmark Description: Use a food web to identify and distinguish producers, consumers, and decomposers. Explain the pathway of energy transfer through trophic levels and the reduction of available energy at successive trophic levels.

28. Gene mutations may cause changes to phenotypes. Such mutations can be inherited. Which of the following must occur for a gene mutation to cause a phenotypic change that is passed on to the next generation?

- Ⓐ The mutation must be caused by radiation.
- Ⓑ The mutation must affect the DNA of a gamete.
- Ⓒ The mutation must affect the DNA of a somatic cell.
- Ⓓ The mutation must be caused by a random replication error.

Answer Key: B

Percentage of Students Answering Correctly: 65%

Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.L.16.4

Benchmark Description: Explain how mutations in the DNA sequence may or may not result in phenotypic change. Explain how mutations in gametes may result in phenotypic changes in offspring.

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29. At specific times in the cell cycle, cells contain chromosomes with sister chromatids. Mistakes sometimes happen during the cell cycle, and the sister chromatids do not separate from each other. Which of the following **best** explains how an improper separation of sister chromatids can lead to phenotypic changes in an organism?

- Ⓐ The chromosomes in daughter cells have unequal numbers of mutations.
- Ⓑ The chromosomes in daughter cells do not contain centromeres.
- Ⓒ Daughter cells cannot use the chromosomes during replication.
- Ⓓ Daughter cells contain different numbers of chromosomes.

Answer Key: D

Percentage of Students Answering Correctly: 45%

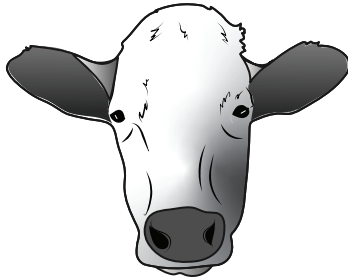
Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.L.16.4

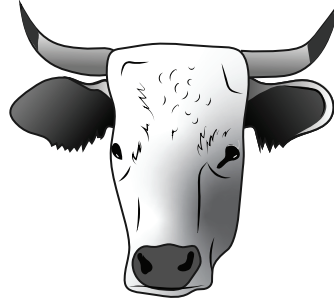
Benchmark Description: Explain how mutations in the DNA sequence may or may not result in phenotypic change. Explain how mutations in gametes may result in phenotypic changes in offspring.

- 30.** In some breeds of cattle, the presence of horns is determined by a single gene with two alleles, and the type of hooves is determined by a different gene with two alleles. The illustrations below represent the two phenotypes for each trait. These illustrations are not to scale.

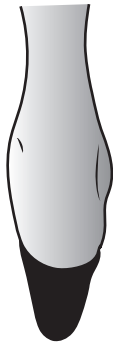
No Horns



Horns



Mule Hoof



Cloven Hoof



Scientists conducted two crosses between male and female cattle with different phenotypes. The crosses and the approximate percentage of each type of offspring from each cross are listed in the table below.

RESULTS OF CATTLE CROSSES

Cross 1		
Male	Female	Offspring
• Heterozygous for no horns • Heterozygous for cloven hooves	• Homozygous for horns • Homozygous for mule hooves	25% no horns and cloven hooves
		25% no horns and mule hooves
		25% horns and cloven hooves
		25% horns and mule hooves
Cross 2		
Male	Female	Offspring
• Homozygous for horns • Homozygous for mule hooves	• Heterozygous for no horns • Heterozygous for cloven hooves	25% no horns and cloven hooves
		25% no horns and mule hooves
		25% horns and cloven hooves
		25% horns and mule hooves

Which of the following **best** describes the inheritance pattern of the traits?

- Ⓐ Having horns and having mule hooves are both autosomal recessive.
- Ⓑ Not having horns and having cloven hooves are both sex-linked dominant.
- Ⓒ Having horns is autosomal recessive, but having mule hooves is sex-linked recessive.
- Ⓓ Not having horns is autosomal dominant, but having cloven hooves is autosomal recessive.

Answer Key: A

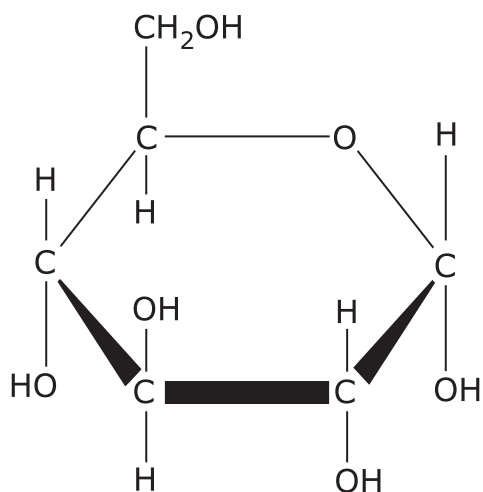
Percentage of Students Answering Correctly: 33%

Reporting Category: Classification, Heredity, and Evolution

Benchmark: SC.912.L.16.2

Benchmark Description: Discuss observed inheritance patterns caused by various modes of inheritance, including dominant, recessive, codominant, sex-linked, polygenic, and multiple alleles.

31. As part of an investigation, a student drew the structure shown below.



What type of biological molecule did the student draw?

- Ⓐ a fatty acid
- Ⓑ a nucleotide
- Ⓒ an amino acid
- Ⓓ a monosaccharide

Answer Key: D

Percentage of Students Answering Correctly: 39%

Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.L.18.1

Benchmark Description: Describe the basic molecular structures and primary functions of the four major categories of biological macromolecules.

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- 32.** Scientists have determined that approximately 20 percent of the blood in the human body flows to muscles when at rest. Which of the following **best** explains how blood flow is expected to change during exercise?
- Ⓐ Exercise increases the heart rate, which causes more blood to flow to the muscles.
 - Ⓑ Exercise increases blood pressure, which decreases the rate of blood flow to the muscles.
 - Ⓒ Exercise causes blood vessels to contract, which speeds up the flow of blood to the muscles.
 - Ⓓ Exercise results in a loss of water, which slows the amount of blood that flows to the muscles.

Answer Key: A

Percentage of Students Answering Correctly: 78%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.L.14.36

Benchmark Description: Describe the factors affecting blood flow through the cardiovascular system.

- 33.** A student compared the structures of plant cells and animal cells. After creating a list of similarities and differences between the cells, the student wrote the sentence below.

Animal cells have mitochondria, but plant cells do not.

Which statement **best** evaluates this claim?

- Ⓐ Correct—animal cells have mitochondria, and plant cells do not need to break down glucose.
- Ⓑ Correct—animal cells have mitochondria, and plant cells use chloroplasts to break down glucose.
- Ⓒ **Incorrect—plant cells and animal cells have mitochondria because both cell types need to break down glucose.**
- Ⓓ Incorrect—plant cells have mitochondria and animal cells do not because animal cells do not need to break down glucose.

Answer Key: C

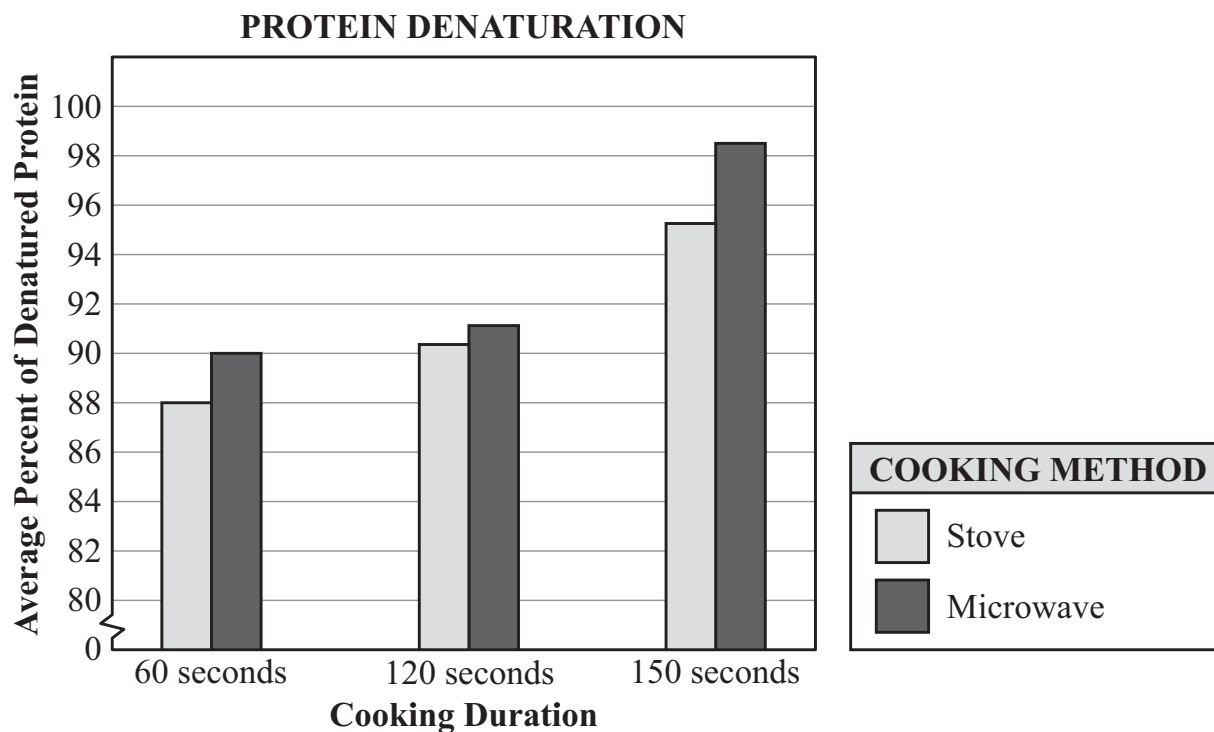
Percentage of Students Answering Correctly: 55%

Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.L.14.3

Benchmark Description: Compare and contrast the general structures of plant and animal cells. Compare and contrast the general structures of prokaryotic and eukaryotic cells.

34. Cooking eggs denatures the structure of the protein in egg whites. When the protein is denatured, the egg white changes appearance from clear to white. A student investigated how cooking egg whites on a stove and cooking egg whites in a microwave affected the amount of protein that was denatured. The student used an instrument to measure the percent of denatured protein in a sample. Each cooking method was repeated on numerous samples, and the average measurements are shown in the graph below.



Which of the following is the **best** analysis of these data?

- (A) The amount of protein that appears to be denatured in egg whites was less in the microwave than on the stove.
- (B) Similar amounts of protein appear to be denatured in egg whites cooked on the stove for 120 seconds and for 150 seconds.
- (C) Less protein appears to be denatured in the egg whites cooked for 150 seconds than in those cooked for shorter durations of time.
- (D) More protein appears to be denatured in egg whites cooked in the microwave between 120 seconds and 150 seconds than in those cooked on the stove.

Answer Key: D

Percentage of Students Answering Correctly: 74%

Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.N.1.1

Benchmark Description: Define a problem based on a specific body of knowledge, for example: biology, chemistry, physics, and earth/space science, and do the following:

1. pose questions about the natural world,
2. conduct systematic observations,
3. examine books and other sources of information to see what is already known,
4. review what is known in light of empirical evidence,
5. plan investigations,
6. use tools to gather, analyze, and interpret data (this includes the use of measurement in metric and other systems, and also the generation and interpretation of graphical representations of data, including data tables and graphs),
7. pose answers, explanations, or descriptions of events,
8. generate explanations that explicate or describe natural phenomena (inferences),
9. use appropriate evidence and reasoning to justify these explanations to others,
10. communicate results of scientific investigations, and
11. evaluate the merits of the explanations produced by others.

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- 35.** Horses can have many different shades of coat colors. One of the basic coat colors of horses is red. An additional gene, with a mutation called cream, modifies the red color. The chart below shows the effect that different numbers of the cream mutation have on the coat color of a red horse.

EFFECT OF CREAM MUTATION ON COAT COLOR

Basic Coat Color	Number of Cream Mutation Alleles	Final Coat Color
Red	0	Red
Red	1	Palomino
Red	2	Cremello

If two palomino horses mate, what percentage of offspring are expected to be cremello in color?

- Ⓐ 0%
- Ⓑ 25%
- Ⓒ 50%
- Ⓓ 75%

Answer Key: B

Percentage of Students Answering Correctly: 45%

Reporting Category: Classification, Heredity, and Evolution

Benchmark: SC.912.L.16.1

Benchmark Description: Use Mendel's laws of segregation and independent assortment to analyze patterns of inheritance.

36. A student uses a classroom microscope to observe the cells of an organism. The student claims that the cells are from a plant. Which of the following student observations provides evidence to support the claim?

- Ⓐ Each of the cells contains cytoplasm.
- Ⓑ Each of the cells contains chloroplasts.
- Ⓒ None of the cells contain mitochondria.
- Ⓓ None of the cells contain a central vacuole.

Answer Key: B

Percentage of Students Answering Correctly: 72%

Reporting Category: Molecular and Cellular Biology

Benchmark: SC.912.L.14.3

Benchmark Description: Compare and contrast the general structures of plant and animal cells. Compare and contrast the general structures of prokaryotic and eukaryotic cells.

Biology 1 Test Release Support Document

- 37.** Between 1945 and 1952, the brown tree snake was accidentally introduced to Guam, a small island in the Pacific Ocean. This snake population has increased rapidly since its introduction. With a growing number of these snakes on Guam, many native bird species have become extinct and other native organisms are threatened. Which of the following inferences can **best** be supported by these observations?
- Ⓐ The number of nonnative predators may decrease when competition between predatory species is absent in an ecosystem.
 - Ⓑ Introducing a nonnative species to an isolated island may increase the biodiversity of natural predators found on that island.
 - Ⓒ Introducing a nonnative species into an ecosystem may cause a decrease in biodiversity of other species that are without natural predators.
 - Ⓓ The number of native predators may increase when nonnative species compete with natural predators for resources on an isolated island.

Answer Key: C

Percentage of Students Answering Correctly: 70%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.N.1.6

Benchmark Description: Describe how scientific inferences are drawn from scientific observations and provide examples from content being studied.

38. Groups of students investigated the effect of detergents on algal growth. Each group used the procedure listed below to compare detergents with and without phosphates.

1. Label two 1000-milliliter (mL) beakers 1 and 2, then add 500 mL of pond water that contains algae to each beaker.
2. Use separate, clean pipettes to collect a 1 mL sample from each beaker, then place each sample on a separate slide.
3. Use a compound microscope to observe the algae, then shade in how much of the field of view has algal growth.
4. Add 10 drops of detergent containing phosphate to beaker 1. Mix gently.
5. Add 10 drops of detergent that does not contain phosphate to beaker 2. Mix gently.
6. Place both beakers near a sunny window.
7. Each day for 5 days, collect a 1 mL sample from each beaker, then shade in the field of view to indicate the estimated final algal growth.
8. Compare the results with those of other groups of students.

Which of the following describes the **most** significant flaw in this investigation?

- Ⓐ In the investigation, the beakers were not kept near a sunny window for a long enough time period.
- Ⓑ The investigation does not include a beaker that contains pond water without any detergent.
- Ⓒ The investigation does not include a beaker that contains different types of algae.
- Ⓓ In the investigation, the beakers were not maintained at the same temperature.

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Answer Key: B

Percentage of Students Answering Correctly: 61%

Reporting Category: Organisms, Populations, and Ecosystems

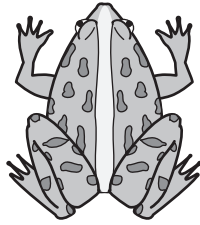
Benchmark: SC.912.N.1.1

Benchmark Description: Define a problem based on a specific body of knowledge, for example: biology, chemistry, physics, and earth/space science, and do the following:

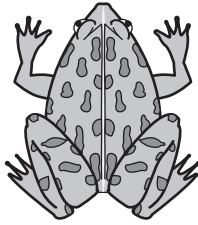
1. pose questions about the natural world,
2. conduct systematic observations,
3. examine books and other sources of information to see what is already known,
4. review what is known in light of empirical evidence,
5. plan investigations,
6. use tools to gather, analyze, and interpret data (this includes the use of measurement in metric and other systems, and also the generation and interpretation of graphical representations of data, including data tables and graphs),
7. pose answers, explanations, or descriptions of events,
8. generate explanations that explicate or describe natural phenomena (inferences),
9. use appropriate evidence and reasoning to justify these explanations to others,
10. communicate results of scientific investigations, and
11. evaluate the merits of the explanations produced by others.

- 39.** In the African grass frog, the expression of multiple genes, including a gene called *ASIP*, is responsible for three different spinal stripe phenotypes. The illustration below shows frogs with the three observed phenotypes.

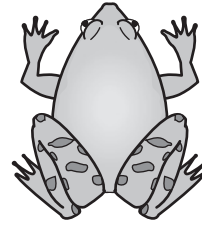
AFRICAN GRASS FROG SPINAL STRIPE PHENOTYPES



Wide Stripe



Thin Stripe



No Stripe

Scientists observed and counted frogs living on land surfaces and frogs living in trees. The scientists claim that the spinal stripe may be selected for when frogs live on land surfaces and selected against when frogs live in trees. How should this claim be evaluated?

- Ⓐ Scientists should measure the spinal stripe width and calculate the average width for land-dwelling frogs and for tree-dwelling frogs.
- Ⓑ Scientists should confirm that the frogs living on land surfaces and the frogs living in trees express the *ASIP* gene from the same chromosomal location.
- Ⓒ Scientists should determine whether there is a difference in the frequency of expression of the *ASIP* gene between land-dwelling frogs and tree-dwelling frogs.
- Ⓓ Scientists should compare species of frogs living on land surfaces and species of frogs living in trees to determine whether other populations have a spinal stripe.

Answer Key: C

Percentage of Students Answering Correctly: 46%

Reporting Category: Classification, Heredity, and Evolution

Benchmark: SC.912.N.1.3

Benchmark Description: Recognize that the strength or usefulness of a scientific claim is evaluated through scientific argumentation, which depends on critical and logical thinking, and the active consideration of alternative scientific explanations to explain the data presented.

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40. Genetic drift can result in fewer types of phenotypes among a species.
Which of the following describes how genetic drift can result in this change?

- ☒ (A) It reduces genetic variation.
- ☐ (B) It decreases the size of the chromosomes.
- ☐ (C) It reduces the number of genes translated on the ribosomes.
- ☐ (D) It decreases the number of genes transcribed in the nucleus.

Answer Key: A

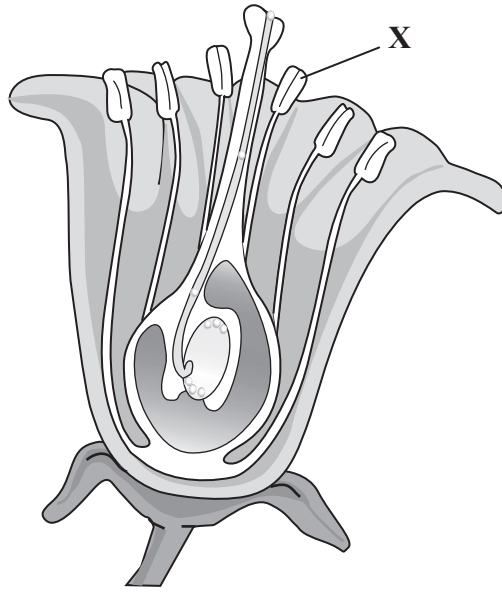
Percentage of Students Answering Correctly: 60%

Reporting Category: Classification, Heredity, and Evolution

Benchmark: SC.912.L.15.14

Benchmark Description: Discuss mechanisms of evolutionary change other than natural selection such as genetic drift and gene flow.

41. Shown below is a cross section of a flower with one part labeled X.



How is the structure of part X related to its function?

- Ⓐ Encasing the egg, it provides protective storage.
- Ⓑ Its green color provides for efficient light absorption.
- Ⓒ Its oval shape attracts birds searching for bugs or seeds.
- Ⓓ **Protruding from the surface, it is accessible to pollinators.**

Answer Key: D

Percentage of Students Answering Correctly: 78%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.L.14.7

Benchmark Description: Relate the structure of each of the major plant organs and tissues to physiological processes.

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42. After fertilization occurs, a human embryo requires multiple months to complete fetal development. Which statement describes an important function of the amniotic fluid during fetal development?

- Ⓐ It supplies oxygen to the fetus.
- Ⓑ It protects the fetus from injury.
- Ⓒ It transfers nutrients to the fetus.
- Ⓓ It separates the fetus from waste.

Answer Key: B

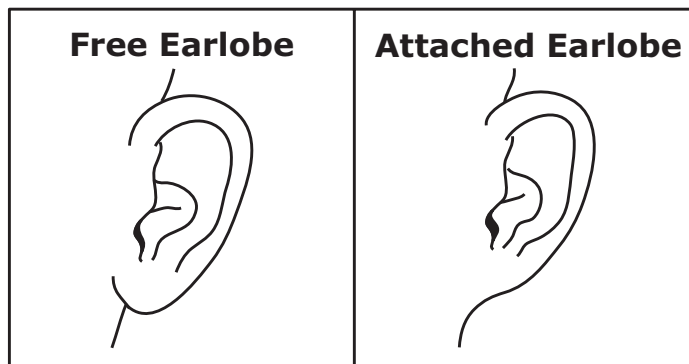
Percentage of Students Answering Correctly: 33%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.L.16.13

Benchmark Description: Describe the basic anatomy and physiology of the human reproductive system. Describe the process of human development from fertilization to birth and major changes that occur in each trimester of pregnancy.

- 43.** Some people have classified human earlobes as either being free or attached. Two sets of researchers investigated the inheritance of earlobe type by categorizing a large number of offspring from many sets of parents. The earlobe types and the results of the two investigations are shown below.



FREE vs. ATTACHED EARLOBES

Parent 1/Parent 2	Investigation 1			Investigation 2		
	Number of Offspring		Percentage with Free Earlobes	Number of Offspring		Percentage with Free Earlobes
	Free Earlobes	Attached Earlobes		Free Earlobes	Attached Earlobes	
Free/Free	13	1	93	11	21	34
Free/Attached	9	9	50	68	111	38
Attached/Attached	8	34	19	35	90	28

Which of the following is a correct analysis of these data?

- Ⓐ The percentage of offspring with free earlobes is similar for all sets of parents in the investigations.
- Ⓑ A comparison of Investigation 1 and Investigation 2 shows that offspring of two parents with attached earlobes are most likely to have free earlobes.
- Ⓒ A comparison of the investigations shows that the largest difference in the percentage of offspring with free earlobes occurs when both parents have free earlobes.
- Ⓓ The percentage difference between offspring having free earlobes in the investigations is greater when the parents have different types of earlobes than when they have the same type of earlobes.

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Answer Key: C

Percentage of Students Answering Correctly: 51%

Reporting Category: Classification, Heredity, and Evolution

Benchmark: SC.912.N.1.1

Benchmark Description: Define a problem based on a specific body of knowledge, for example: biology, chemistry, physics, and earth/space science, and do the following:

1. pose questions about the natural world,
2. conduct systematic observations,
3. examine books and other sources of information to see what is already known,
4. review what is known in light of empirical evidence,
5. plan investigations,
6. use tools to gather, analyze, and interpret data (this includes the use of measurement in metric and other systems, and also the generation and interpretation of graphical representations of data, including data tables and graphs),
7. pose answers, explanations, or descriptions of events,
8. generate explanations that explicate or describe natural phenomena (inferences),
9. use appropriate evidence and reasoning to justify these explanations to others,
10. communicate results of scientific investigations, and
11. evaluate the merits of the explanations produced by others.

- 44.** Some viruses enter the human body and attack the immune system. The immune system is then unable to remove these viruses, so they remain active in the body for a long time. Scientists have developed treatments using different types of medications to remove a particular virus from the body. The chart below describes the actions of five types of these medications.

ACTIONS OF MEDICATIONS

Type	Action
1	Turns off a protein that the virus needs to make copies of itself
2	Blocks the virus from entering some types of human immune cells
3	Interrupts the function of a protein that the virus needs to make copies of itself
4	Provides nonfunctional pieces of the building blocks that the virus uses to make copies of itself
5	Disables a protein that the virus uses to insert genetic material into some types of human immune cells

Which statement **best** explains why patients who carry this kind of virus are prescribed a long-term treatment of more than one type of the listed medications?

- Ⓐ Using more than one type of these medications causes the virus to rearrange its genetic material before it is replicated.
- Ⓑ Using more than one type of these medications prevents the virus from copying its genetic material at a faster rate than normal.
- Ⓒ Using more than one type of these medications helps the virus enter the parts of the body that are resistant to one type of medication.
- Ⓓ Using more than one type of these medications prevents the formation of versions of the virus that are resistant to any specific type of medication.

Answer Key: D

Percentage of Students Answering Correctly: 49%

Reporting Category: Organisms, Populations, and Ecosystems

Benchmark: SC.912.L.16.10

Benchmark Description: Evaluate the impact of biotechnology on the individual, society, and the environment, including medical and ethical issues.